ExMortis [beta-1]

The players' manual for ExMortis, a mod for Nitroplus Blasterz: Heroines Infinite Duel (Steam).

This is a public beta. Section 18 lists what is known to be rough, and section 17 says what to send when something goes wrong. Credits, licences and the legal notice are in sections 19 to 21.

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1. What ExMortis adds

- **Rollback netplay** in Player Match, in the game's own menus. The original delay netcode is replaced.
- **Rooms** with a set score (first to N), rotation for three or more players, Practice rooms, quick comments, a join link and Steam or Discord invites.
- **Spectating**: waiting players watch the room's matches live and can join a match that is already running.
- **A Ranked lobby** of its own: challenge a player, play a first-to-2 set, win or lose VP, watch other sets.
- **Replays** of online matches in the game's Replay Theater, with a position bar, speed, frame steps and Take Control.
- **Training tools**: save states, frame step, slow motion, rewind, hitboxes, a frame meter, reversals, longer dummy recordings.
- **Custom colours**: 64 extra colours for every fighter, made in an editor inside the game and shown to other players.
- **Controls**: every keyboard key can be changed, menus have their own keys, pads get menu buttons and L3 / R3.
- **A window you can resize:** drag its edges or corners, maximize it, or use Borderless. Its place and size are remembered. Alt+Enter swaps Windowed and Borderless.
- **A faster start:** press Confirm to skip the startup logos.
- **Fixes** to the original game: an even 60 frames per second, no screen going dark in a long fight, no stall from streamed sound.

Everything is in the game's own menus, in English and Japanese. There is no launcher and no settings program.

The game itself is not included. You need the Steam version.

2. Requirements

- **The game:** Nitroplus Blasterz: Heroines Infinite Duel, the current Steam version. The mod checks the game's file. On any other version its menus, netplay and controls stay off.
- **Windows 10 or 11.**
- **For online play, the same mod file.** Everyone you play with needs the same `dinput8.dll`, byte for byte. Two files with the same version name but from different builds refuse each other. The version name is in the window title: `Nitroplus Blasterz ... +ExMortis [beta-1]`.
- **Players without the mod cannot play with you.** They see your room in the list, but they are asked to leave it.

3. Install, update, uninstall

Install

1. Close the game.
2. In Steam, right-click the game and choose **Manage > Browse local files**.
3. Put `dinput8.dll` next to `Nitroplus Blasterz.exe`.
4. Start the game from Steam as usual.

If a `dinput8.dll` from another mod is already there, the two cannot be used together. Keep a copy of the other file before you replace it.

If the download or the file is blocked

A browser may refuse the download (`This file cannot be downloaded`, `blocked`, `Virus detected`), or an antivirus tool may remove `dinput8.dll`. This is expected for a new mod file. It is a caution about files of this kind, not a finding about this one.

What to do, in order

Downloading, unzipping or restoring a file does not run it. `dinput8.dll` runs only when the game starts with it in the game's folder. So get the file first, check it, and install it last:

1. Tell the browser to keep the download (the steps for each browser are below). In our own tests the zip came through Chrome, Edge and Firefox without a warning. A bare `dinput8.dll` was stopped by Chrome and Edge.
2. Unzip it. If Windows Security removes the zip or the DLL, follow its part below.
3. Check the file's fingerprint (next part).
4. If it matches, install it. If it does not match, delete the file and tell us.

Check that the file is ours

A fingerprint (SHA-256) is a long value worked out from the file's contents. Change one byte of the file and the value is a different one. We post the fingerprint of `dinput8.dll` where we share the download. The package's `SHA256.txt` holds the same value.

To read your file's value, open the folder with the file, click the address bar, type `powershell`, press Enter and run:

```
Get-FileHash dinput8.dll
```

The same value as ours means you have exactly the file we built. A different value means a damaged or changed file: delete it. Go by the posted value, not by the one in the zip, and take the mod only from where we share it.

Why browsers warn

A browser's warning comes mostly from a reputation check, not from a look at what the file does. For a program file (an `.exe`, a `.dll`, or a zip with one inside) the browser asks a reputation service: Safe Browsing for Chrome and Firefox, SmartScreen for Edge and Windows. The service answers from two things:

- **How many people have downloaded this exact file, and for how long.** A file it has rarely or never seen gets `not commonly downloaded` or `could harm your device`. Every build of the mod is a new file and starts at zero. With a small group of testers the count stays low, so the warning can come back at every version.
- **Who signed it.** A signed file carries its publisher's certificate, and trust in the publisher passes on to each new file. A certificate costs money every year. Free mods rarely have one, so each file is judged alone, as unknown.

Chrome also warns about the kind of file alone. A bare `.dll` is a kind people rarely download, so Chrome says `the file type isn't commonly downloaded and it may be dangerous` for any DLL, from anyone.

So the warning means "we do not know this file yet". It does not mean something was found in it.

Why antivirus tools flag it

`dinput8.dll` is the name of a part of Windows that reads keyboards and pads. The game asks Windows for it at startup, and Windows looks in the game's own folder first. Our file is found there and loads into the game. It passes the game's input calls on to the real Windows file, then changes the running game in memory: that is how rollback, the new menus and the rest get in. The game's files on disk stay as they are, which is why deleting `dinput8.dll` brings the original game back. Many game mods work this way.

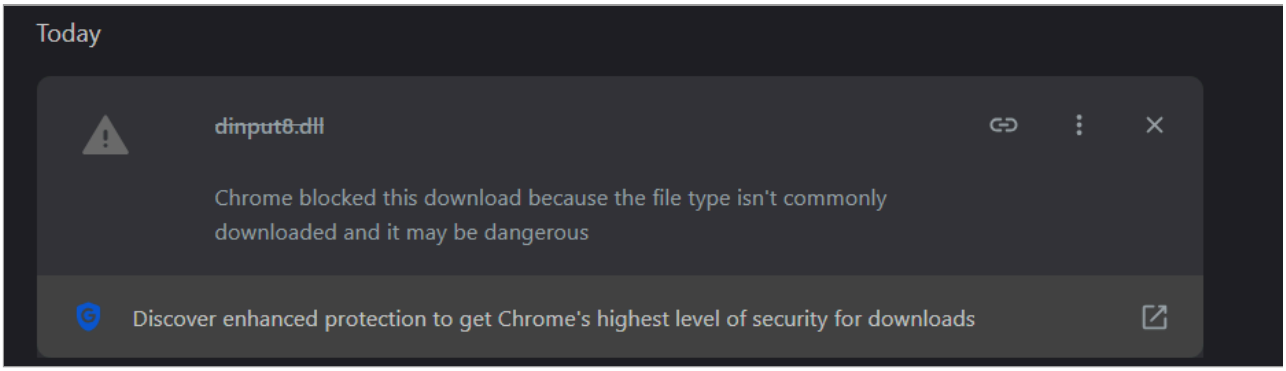
An antivirus tool sees only the shape of that: an unsigned file, named like a Windows file, in a program's folder, changing that program while it runs. Harmful programs use the same trick, so the tool scores the shape, not what the file does. The result is often a general name such as `Trojan:Win32/Wacatac.B!ml`. A name ending in `!ml` usually marks a guess made by a model, not a match with a known threat.

What the mod does on your PC

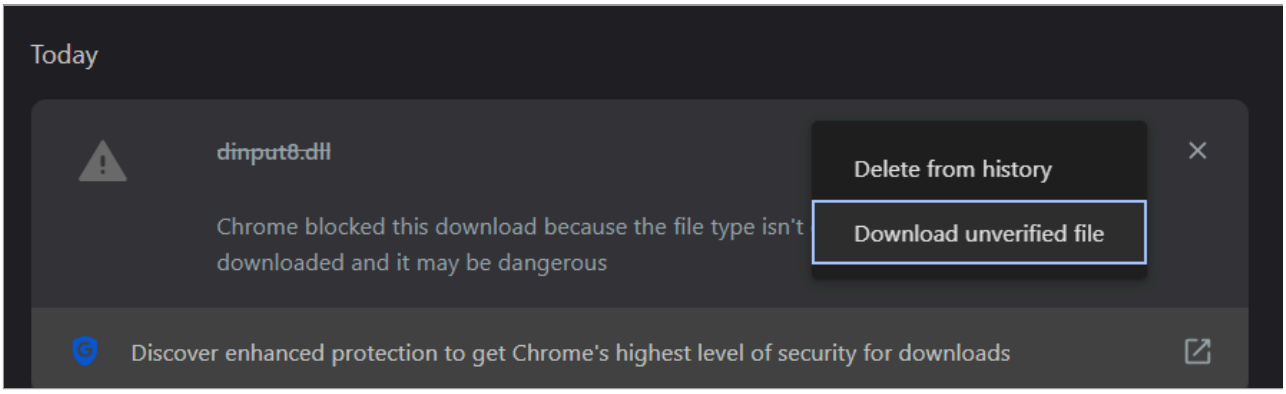
- It runs only inside the game, while the game runs. Close the game and nothing of it is running.
- It needs no administrator rights and never asks for them.
- It installs nothing: no service, no autostart entry, no scheduled task, no firewall rule.
- It keeps its files in the `exmortis` folder next to the game. The few other traces are listed under "Uninstall" below.
- Online play goes through Steam, like the game's own.

Chrome (also Brave, Opera, Vivaldi)

1. Press Ctrl+J. The file is crossed out, with `Chrome blocked this download because the file type isn't commonly downloaded and it may be dangerous`. Other versions say `Suspicious download blocked` or `Dangerous download blocked`.

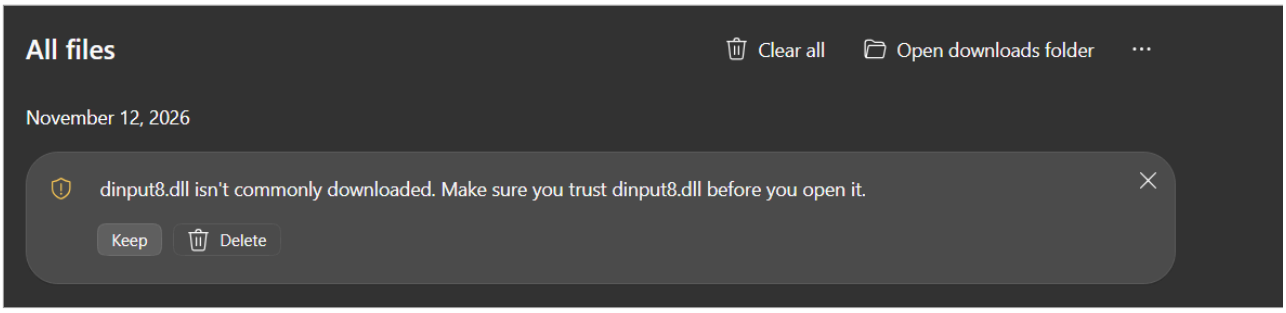


2. Click the three dots on the entry and choose **Download unverified file**. Other versions say **Download suspicious file**, **Download dangerous file** or **Keep**.



Edge

1. Press Ctrl+J. The entry reads `dinput8.dll isn't commonly downloaded. Make sure you trust dinput8.dll before you open it.` Other versions say `was blocked because it could harm your device`.



2. Click **Keep** on the entry. In the small downloads list it is under **...**.
3. If a window opens, click **Show more**, then **Keep anyway**.

Firefox

Firefox took both the zip and the bare DLL without a warning in our tests. If it does show `This file is not commonly downloaded`:

1. Click the downloads arrow at the top right, or press Ctrl+J.
2. Click the arrow next to the file, or right-click it, and choose **Allow download**. Confirm.

`Virus detected` **in any browser**

`Failed - Virus detected` or `Couldn't download - Virus detected` comes from Windows Security, not from the browser: it removed the file as it arrived. See the next part.

Windows Security (Microsoft Defender)

Microsoft Defender found nothing in this build when we scanned it before sharing it. It can still decide differently on your PC, at once or after a later update. You then see one of these:

- a `Threats found` note from Windows Security,
- `dinput8.dll` is gone from the game's folder, and the window title has no `+ExMortis`,
- unzipping stops with `Operation did not complete successfully because the file contains a virus or potentially unwanted software`,
- `Failed - Virus detected` in the browser.

Putting the file back is not enough, because Defender finds it again. Tell it to leave the game's folder alone first. You need an administrator account for this.

1. In Steam, right-click the game and choose **Manage > Browse local files**. That is the game's folder.
2. Open the Start menu, type `Windows Security` and open it.
3. Open **Virus & threat protection**. Under **Virus & threat protection settings**, click **Manage settings**.
4. Scroll down to **Exclusions** and click **Add or remove exclusions**. Click **Add an exclusion**, choose **Folder** and pick the game's folder. Defender no longer checks the files in that folder. This runs nothing.
5. Get the file into that folder: copy the zip into the game's folder and unzip it there. If Defender took the DLL out of the game's folder itself, open **Virus & threat protection > Protection history**, pick the entry and choose **Actions > Restore**. The history keeps entries for about two weeks.
6. Check the fingerprint, then start the game. If it does not match, delete the file and remove the exclusion.

If Defender removes the zip itself before you can unzip it, open **Protection history**, pick the entry, choose **Actions > Allow on device**, and download the zip again. This lets through every file that gets the same threat name, so take it back afterwards under **Virus & threat protection > Allowed threats**.

Other antivirus tools can flag the file for the same reasons. Each has its own list of exceptions and its own quarantine.

What the first launch does

- It makes a folder named `exmortis` next to the game. Everything the mod keeps is in it (section 13).
- It writes `exmortis\exmortis.ini` with the game's default keys. Nothing has to be copied in by hand.
- It copies your save to `exmortis\SaveData.dat` (see "Your save" below).
- The window title now ends with `+ExMortis [beta-1]`. If it does not, the mod is not running (section 17).



Keyboard keys and pad buttons you set in the original game are taken over. The keyboard keys are copied once, on the first launch; after that, change them in Keyboard Config (section 12).

Update

1. Close the game.
2. Replace `dinput8.dll` with the new one.
3. Keep the `exmortis` folder. It holds your settings, replays, colours and rating.

Uninstall

Delete `dinput8.dll`. The original game is back, with your save as it was before the mod first ran.

These things stay until you remove them yourself:

What	Where	Remove it by
The mod's folder	`exmortis` next to the game	Deleting the folder. Your replays, colours and Ranked rating go with it.
A copy of your Ranked rating	Steam Cloud, one small file (`exmortis-giga-rating.dat`)	It is harmless. The original game never reads it.
The Performance Test's record for this PC	Registry, your user only: `Software\ExMortis\Rollback`	Deleting that key in the Registry Editor.
Discord's launch entry	Registry, your user only: `Software\Classes\discord-...`	Deleting that key.
L3 / R3 pad bindings	The game's own pad file	Changing them in Button Config. The original game understands them.

The mod sets no firewall rules, no scheduled tasks and no display modes.

Your save

The mod never writes the game's own save file.

- On the first launch the mod copies the game's save (`SaveData\SaveData.dat`) to `exmortis\SaveData.dat`.
- From then on, while the mod is in, the game reads and writes only that copy.
- When you remove the mod, the game uses its own save again, exactly as it was.

What this means for you:

- What you unlock or change while the mod is in is only in the mod's copy. Steam Cloud does not carry it to another PC.
- What you unlock later without the mod does not flow into the mod's copy. The copy is taken once.
- To start the mod's copy again from the game's save: close the game and delete `exmortis\SaveData.dat`.

4. The first time online: the Performance Test

The first time you open the **Network** menu, a fight starts by itself. This is the Performance Test. It measures how well your PC handles rollback.

- It is an automatic fight of about 40 seconds. The progress shows as `Performance Test N/100`.
- At the end a box shows `Performance OK` or `Test failed`, three scores (`FPS`, `Online`, `CPU`, each out of 10) and one line of advice.
- **It limits nothing.** Every match allows the same 20 frames of rollback whatever the scores are. The test is advice only.
- To skip it, press Start and answer Yes to `Skip the Performance Test?`.
- It runs once per PC and per mod file, so it runs again after an update.
- It only runs with V-Sync off (Options > Display). With V-Sync on it waits, without a message, until the first Network menu visit after you turn V-Sync off.

5. Playing online: Player Match

Everything online starts in the **Network** menu.



Rollback

Every Player Match fight uses rollback netcode. When an input arrives late, the game rewinds a few frames and replays them instead of stopping to wait. There is no switch and no depth setting: every match allows up to 20 frames of rollback, the same for everyone.

During a fight, the bottom line shows `Delay N / Ping Nms`. Options > Game > Delay / Ping hides it.

Messages you may see in a fight:

Message	Meaning
`Waiting for your opponent... (N)`	Their inputs stopped arriving. The game waits N more seconds.
`The connection with your opponent was lost.`	The link dropped. The match ends.
`Your opponent left the match.`	They quit or closed the game.
`The games fell out of sync. The match was stopped.`	The two games disagreed (a desync). Please report it (section 17).

Input Delay

Input Delay is how many frames your own inputs wait before they are used. More delay means less rollback on a slow connection.

Options > Game (GAME OPTION) > Input Delay: `0` to `5`, or `Auto` (the default). The same row is the first row of the Ranked lobby's SETTINGS tab.

- **Auto** picks the delay from your ping at the start of each match.
- **A number** is used as it is. If the connection needs more, it is raised to the lowest value that works when the match starts. If even 5 is not enough, the match plays at 5 and the number is shown in yellow. Expect more rollback then.
- The delay is yours alone. Your opponent has their own.

You can also change the delay without opening Options:

Where	Pad	Keyboard
Room page, the INPUT DELAY block (top right)	LT / RT	`E` / `Q`, or `-` / `=`
Result screen, the Input Delay line	LT / RT	`-` / `=`
During your own online match	(none: LT / RT are fight buttons)	`-` / `=`

A message shows the new value: `Input Delay: N frames`.

Two things to know:

- A delay you step with these keys is **remembered for that opponent** and set again the next time you meet them (up to 64 players).
- It stays with that opponent. Against a player you have no remembered delay for, the Options setting applies again, so Auto comes back by itself.

Create a room

Network > Player Match > Create Room. The rows:

Row	Values
Name	Type it with the keyboard, up to 20 characters. Japanese input works. Enter keeps it, Escape brings the old name back.
Players	2 to 6
Reserved Slots	Seats kept for invited players
Number of Rounds	1 to 5
Connection Status	The game's own connection filter
Room Type	**Match** or **Practice** (section 6)
Password	Optional: 1 to 12 letters and digits (see Room passwords below)

The page opens with your last room's values, except the name. It reminds you: `Players without the mod can't join your room.`



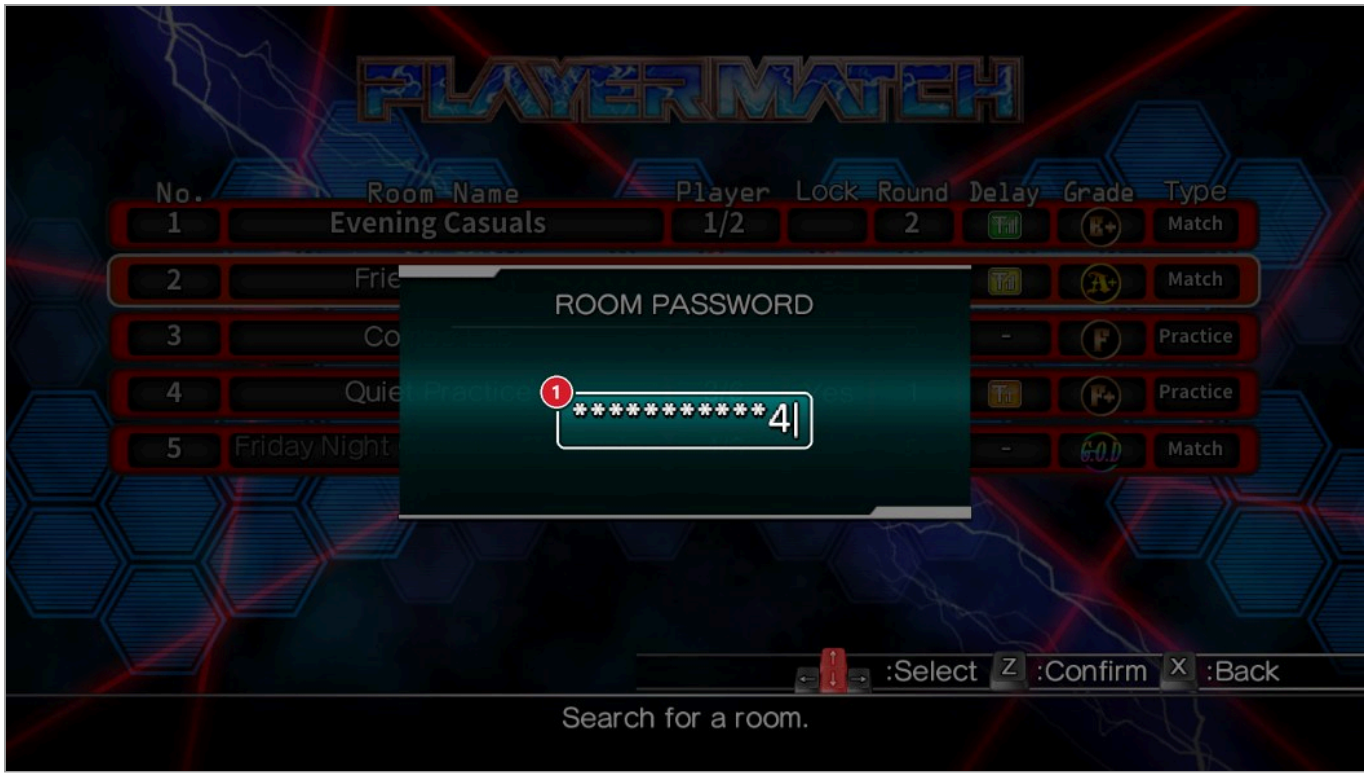
1. Room Type: Match or Practice.
2. Password: `None`, or stars with the last character shown.

Room passwords

A room can have a password, so that only people who know it get in from the room list.

- **Set it** on Create Room's **Password** row, or later in the room with **Room Options > PASSWORD** (host only, between fights). It is 1 to 12 letters and digits. Capital and small letters count as the same. An empty password turns it off.
- **Typing:** use the keyboard and press Enter. With a pad, Up and Down pick the letter, Right adds the next one, Left deletes, Confirm keeps it. Only the last character is shown, the rest are stars.
- **In the room list** a room with a password says `Yes` in the **Lock** column. Confirm on it asks for the password, then joins.
- **Links and invites skip the password.** A join link, a Steam invite, Join Game on a friend and a Discord invite all go straight in. Send those only to people you want in the room.
- **A wrong password** puts you back out with `Wrong password.` You can try again.
- Players who are already in the room stay when the host sets or changes the password.
- **To read it again,** open Room Options and move to PASSWORD: the row shows the password while the cursor is on it, for the host and for the players in the room.
- The password is not kept: your next room starts without one.

It keeps random players out. It is not strong protection.



1. The password as you type it.

The room list

Network > Player Match > Search Room opens the search conditions. Set them, then choose **Search**.

Row	Values
Number of Rounds	`Any`, or 1 to 5
Connection Status	`Anybody`, `3 or More`, `4 or More`
Password	`Any`, or `None`: only rooms without a password
Rooms	`This Version`: rooms made with your build of the mod. `All`: rooms of other builds and of the original game too

The `Searching` box stays up until the listed hosts have answered, three seconds at most.



1. **Lock:** `Yes` when the room has a password.
2. **Delay:** the connection to the host as bars, `-` while it is not measured yet.
3. **Type:** `Match` or `Practice`, `-` for a room you cannot join.

A grey row is a room of another build or of the original game. You cannot join it. Confirm says why.

The list shows seven rooms at a time and follows the cursor when there are more.

The room page

- **The two Stage cards** show P1 and P2: fighter, assists, READY, connection bars, wins and losses in this room, and the set score.
- **The queue** below lists who plays next, in order.
- **The header** shows ROOM, RESERVED, ROUNDS, TIME, QUALITY and TOP VP (the host's Ranked VP).
- **FIRST TO** (top left) is the set length. **INPUT DELAY** (top right) is your delay.
- **One status line** under the queue shows the rollback link while it comes up: `ROLLBACK: CHECKING...`, then `ROLLBACK: CONNECTING...`. READY waits until the link is good.

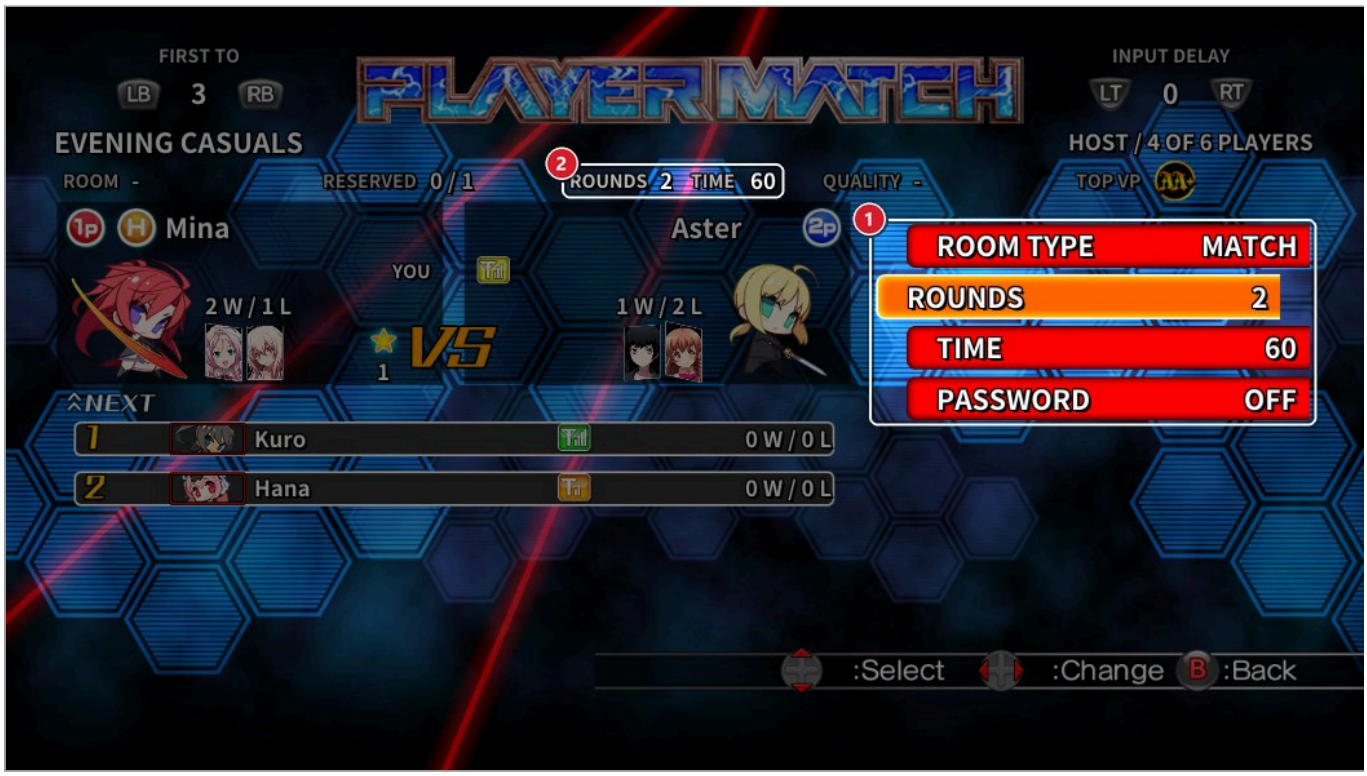


1. FIRST TO: how many wins end the set. The host changes it with LB / RB.
2. INPUT DELAY: your own delay. Change it with LT / RT.
3. The two fighters: characters, partners, connection, wins and losses in this room.
4. The command list.
5. The rollback link's status. READY waits while this line shows a problem.

The command list:

Row	What it does
READY	Ready to fight. For a waiting player while the pair fights, this row is **Spectate**. Alone in the room it is **Training Mode**.
PASS	Give up your turn. For a waiting guest in a room of three or more, this row is **Training Mode**.
SET A COMMENT	A quick comment shown on your row for a few seconds. The game's ten, plus `GG`, `One more?`, `Thanks`, `Brb`.
ROOM OPTIONS	ROOM TYPE (Match / Practice), ROUNDS, TIME and PASSWORD. Anyone can look. Only the host changes them, and only between fights.
PLAYER DATA	The selected player's card.
INVITE TO ROOM	Host only: Steam's invite window.
KICK / LEAVE	The host kicks, a guest leaves.





1. ROOM TYPE, ROUNDS, TIME and PASSWORD (a guest reads On or Off).
2. The room's rounds and time in the header.

More about some rows:

- **FIRST TO** is `1` to `9`, or `-` for no limit. It is the game's Win Limit under a clearer name.
- **ROUNDS** is 1 to 5 rounds to win a game. **TIME** is the round clock: 30, 60 or 99. Both are the whole room's and count from the next fight. A new room starts at 99. A Practice room has neither.

- **Join link:** press Reset on the command list (Left Shift, or Share / Back on a pad). `Join link copied.` appears. Paste the link in a chat. A friend who opens it lands in your room, even if their game was closed.
- **Kick:** a kicked player is put out again each time they come back to that room. A fighter cannot be kicked during their match: `A fighter can't be kicked during a match.`
- **Leave:** a host who is not one of the two fighters cannot leave through the menu while the pair is in a match: `You can leave when the match ends.`
- **Taskbar flash:** if the game is behind other windows, its taskbar button flashes when someone joins, when it is your turn and when your opponent is ready.
- **Preferred Character:** set it on Player Data (Network menu or main menu, fourth row) or in Ranked SETTINGS > CHARACTER. Others see it on the room page before you pick.
- A player whose game stopped responding is removed from the room and told so.

Sets and rotation

- **Two players:** you keep playing each other. The sides never swap.
- **Three or more, with FIRST TO N:** the pair plays until one of them has N wins. The set winner then goes to the back of the queue and everyone returns to the room page.
- **Three or more, with `-`:** the loser leaves the pair after every game, as in the original game.
- The **Wins** counter in the fight and on the Stage cards is the set score. Rematch and Character Selection keep it. Going back to the room page resets it.



1. The set score.
2. The queue: who plays next.

While you wait your turn

- **Spectate** (in READY's place) appears while the pair is fighting. See section 7.
- **Training Mode** (in PASS's place) lets a waiting guest practise in a room of three or more. You come back by yourself when it is your turn. The room's host does not get this row.

After a match

Both players choose **Rematch**, **Character Selection** or **Return**. You have 30 seconds.

Choices	Result
Both pick Rematch	Rematch with the same fighters
Rematch and Character Selection, or both Character Selection	Character Selection
Either picks Return, or nobody chooses in time	Back to the room page

- Character Selection goes straight to the selection screen. Back there undoes one pick. It does not leave.
- Under the rows are a **Save Replay** line (section 10) and the **Input Delay** line.
- When the set is over, the choices are not offered and the screen says `Win Limit reached: returning to room`.

When a room refuses you

You see	Why
`This room's host is not using the mod. Connection failed.`	The room was made without ExMortis.
`This room's host has mod version X. You have Y. Connection failed.`	Different mod files. If both names are the same, four extra characters tell the builds apart: get the same file.
`ROLLBACK: your opponent has version X, you have Y` on the room page	The same, seen from inside a room. READY is held.
`ROLLBACK: no reply from your opponent's game`	Their game did not answer. Wait a moment or rejoin.

If a friend's room is missing from the room list, the two of you do not have the same file. Search with **Rooms** set to `All` to see it: Confirm on the grey row names both versions.

6. Practice rooms

A Practice room is a Player Match room for training together online.

To make one: Create Room > Room Type > **Practice**. In an existing room the host can switch with Room Options > ROOM TYPE, between fights. The room page marks it `PRACTICE` and the room list says `Practice` in its Type column.

In a Practice fight:

- There is no round timer.
- After each combo both fighters get back full health, power and partner meter, each while its option is on.
- **Start opens a pause window over the running fight**, on your PC only. Your fighter stands still while it is open. The fight does not stop.

Pause row	Who	What
Recover Health	Creator, or 1P	Refill health after a combo
Power Meter	Creator, or 1P	Refill the power meter
Partner Meter	Creator, or 1P	Refill the partner meter
Show Hitboxes	You	Draw hitboxes on your screen only (section 9)
Display	You	A list for your screen only: Frame Meter, Gauge Values, Distance, Blast Time, Move Names
Return to Room	Either fighter	End the fight for both and go back to the room page. No result.
Leave	You	Leave the room

Good to know:

- A Practice fight saves no replay and gives no achievements. Nothing goes to Player Data.
- With Recover Health off, a knockout ends the fight like a normal match: the result screen shows and the room counts the win.
- The three refill options belong to the room's creator while the creator is one of the two fighters. When the creator waits, 1P changes them. The other fighter reads them grey.
- In a room of three or more, the fighter who presses **Return to Room** gives up the turn: the next waiting player comes in.
- A creator who is waiting can still leave. If you are watching the fight, leave the watch first (Start). On the room page, Leave ends the pair's fight (`The host ended the fight.`), then asks as always.
- Watchers see the fight with the same rules. A watch stops after about 15 minutes of one fight.

7. Spectating

Up to four waiting players can watch a room's matches live.

- **Automatic:** with Options > Game > **Watch Matches** on (the default), your game follows each match while you wait and brings you back to the room page afterwards.
- **By hand:** with Watch Matches off you stay on the room page. The **Spectate** row still works while the pair is fighting.
- **Joining late:** entering a room during a match, or pressing Spectate, shows `Joining the match in progress...` with a progress bar. Cancel keeps you in the room.
- **Falling behind:** if your game drops more than about 10 seconds behind, it jumps ahead: `Catching up with the match...`.
- **Leaving:** press Start and answer Yes to `Are you sure you want to leave spectating?`. You go back to the room page.
- The watch follows the pair through Rematch and Character Selection, and returns to the room page when they do.
- You see the players' custom colours.

A watch can end with a short reason: the players went back to the room, the match is too long to watch, the connection to the match was lost, four people are already watching, or this PC could not keep up with the match.

Watchers never affect the players. A watcher sends nothing into the fight. There is one small exception: at the end of a set, the next fight's start waits up to about 10 seconds for watchers who are still on their way back to the room.

8. Ranked

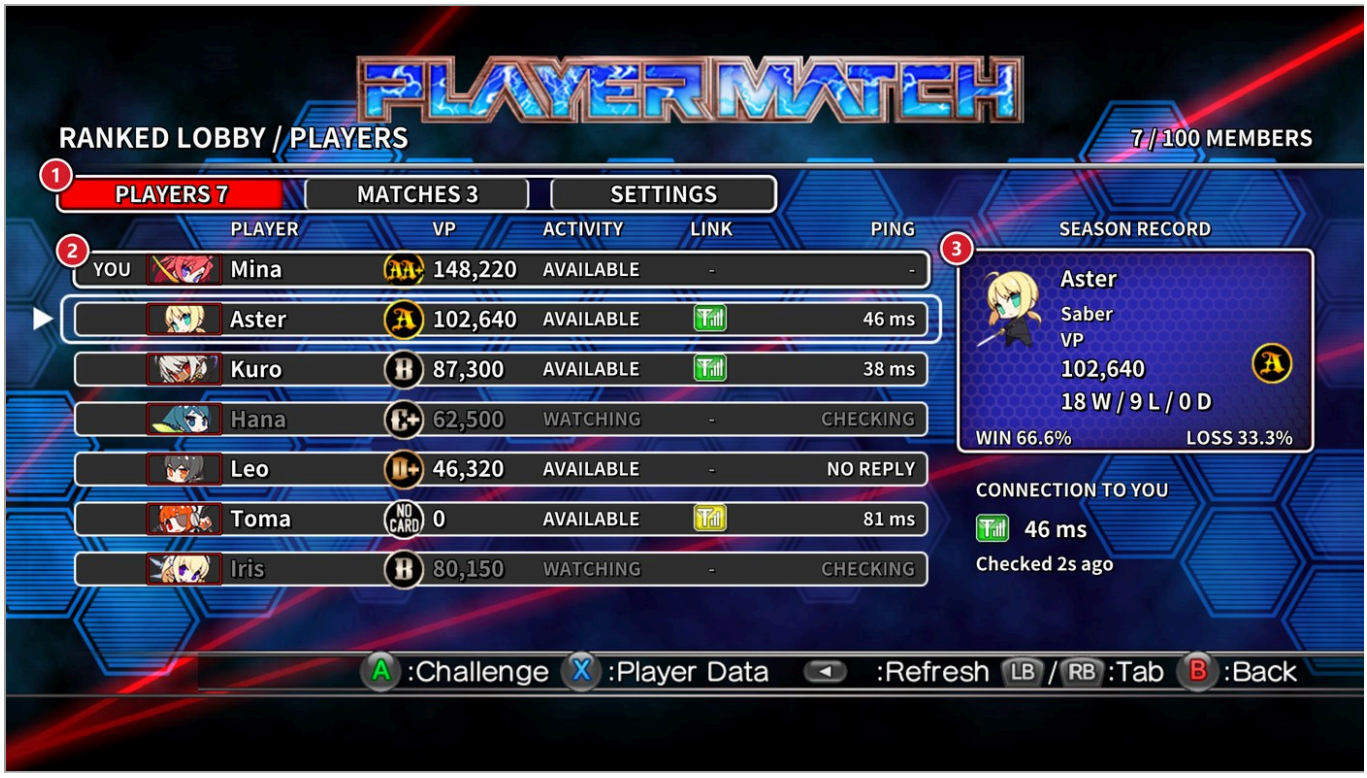
Network > RANKED MATCH opens the ExMortis Ranked lobby. It takes the place of the game's own Ranked Match while the mod is in.

Network > RANKING is unchanged: it is still the game's own leaderboard.

The lobby

One lobby holds everyone on your mod build, up to 100 players. It has three tabs. Change tab with LB / RB.

Tab	Shows	Keys
PLAYERS	Everyone who is free	Confirm: challenge (or Training Mode on your own row). X: Player Data. Reset: shuffle the list on your screen. Back: leave the lobby.
MATCHES	The sets being played, with game number, watchers and both lineups	Confirm: watch. X: Player Data.
SETTINGS	Your Ranked settings	Up / Down, Left / Right. A change saves by itself.



1. The three tabs.
2. Your own row. Confirm here starts Ranked Training.
3. The selected player's card, season record and connection to you.

Players in a set are on MATCHES, not on PLAYERS. Players on another mod build are not listed at all. If you are alone, the list may read `N on another mod version.`

The SETTINGS rows:

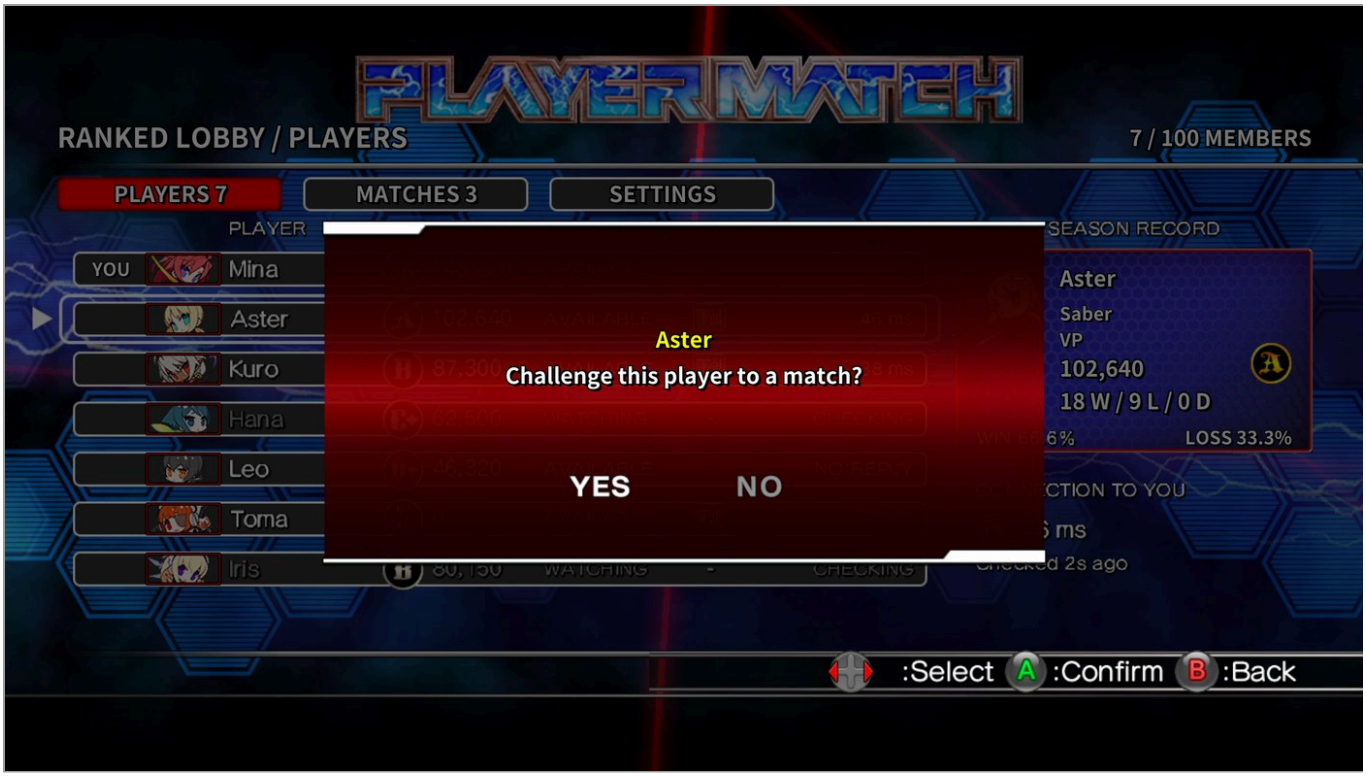
Row	Values	What it does
Input Delay	0 to 5, Auto	The same setting as in Options > Game
Nicknames	On, Off	Show or hide player names in the fight
Allow Spectating	On, Off	Off: your sets are not listed on MATCHES and take no watchers
Max Ping	Off, 50 to 300 ms	Players over it are grey. They cannot challenge you and you cannot challenge them.
Character	A fighter	Your Preferred Character, shown on your row
Sort by Rank	On, Off	Order PLAYERS by VP, with a rank number



Challenge a player

1. On PLAYERS, press Confirm on a free player. `Challenge this player to a match?` Choose Yes.
2. You see `Waiting for an answer...`. Back takes the challenge back.
3. They see `This player challenges you. Accept?`. With no answer in 20 seconds the challenge is withdrawn.
4. If they accept, both games show `Matching...` and go straight to Character Selection. There is no room page and no Ready.

If two players challenge each other at the same moment, the match simply starts.



Ranked Training

Press Confirm on **your own row** to practise while you wait. You stay listed as free. A challenge takes you straight into the set, with no question. The pause menu's return row goes back to the lobby.

A set

- A set is **first to 2 games**. Each game is 2 rounds to win. You keep the same fighters.
- After a game that does not end the set, both players see **Rematch** and **Back to Ranked Lobby**. The next game needs both on Rematch.
- A draw counts for nobody.
- At the end of the set: **Go to Player Room** (both must pick it: a normal Player Match room for the two of you, with no rating) or **Back to Ranked Lobby**.

Watching a set



On MATCHES, press Confirm on a set. You join as a watcher, at the next game or in the one that is running. It works with Watch Matches off. Every way out brings you back to the lobby.

Rating: VP and grade

The lobby keeps its own VP and grade for you, using the game's Ranked rules. Your original Ranked VP is never changed.

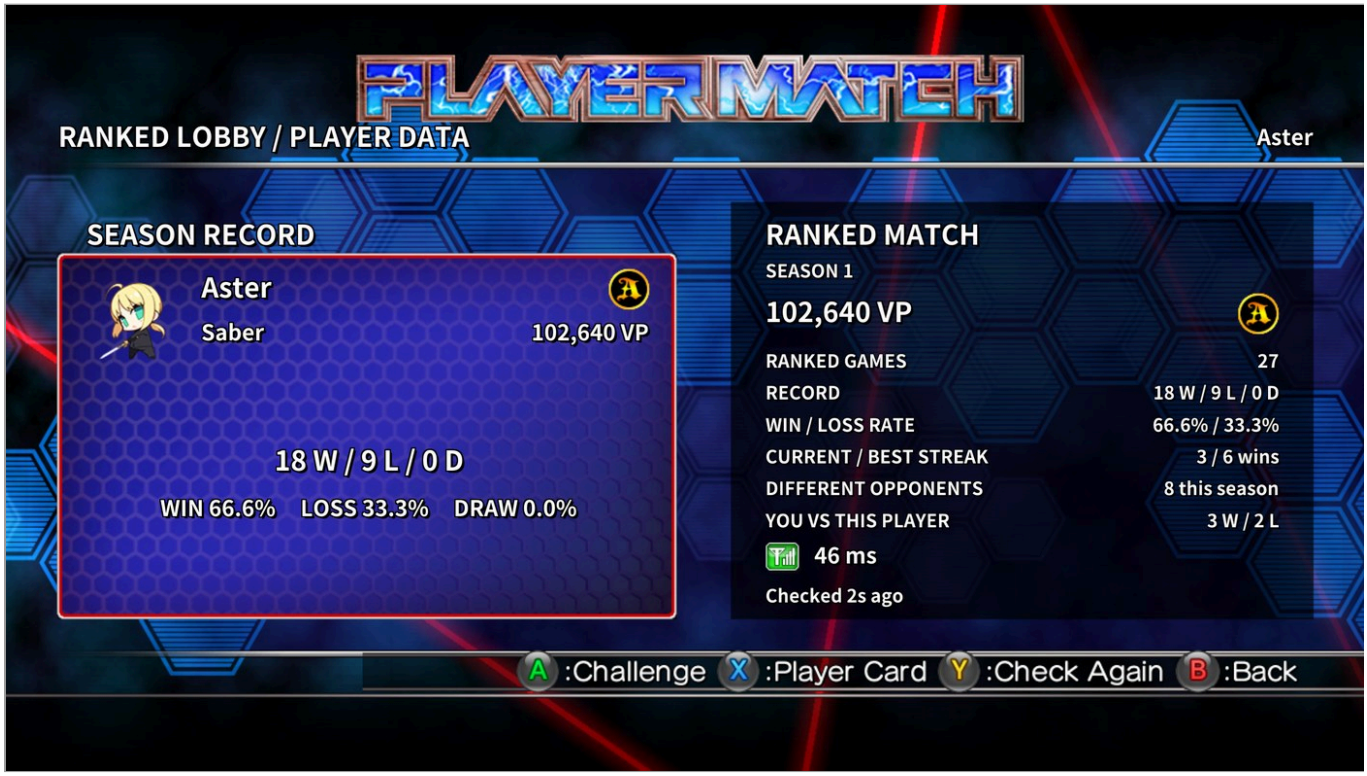
- **Your first rating** is your best original Ranked VP of any fighter, halved. It is set once.
- **What counts:** every game that both PCs confirmed. Unfinished games and Player Room games count for nobody.
- **Set bonus:** winning a game in a set gives +1,000 VP, or +2,000 if you also won the game before it in the same set.
- **Same opponent:** your first four wins in a row against the same player count in full. After that each one counts less, down to 10% from the 13th. The loser's loss shrinks by the same share. Only games of the last 24 hours count toward this, so after a day it starts fresh.
- **Seasons** are calendar months. The first time you open the lobby or Player Data in a new month, your VP is halved and your grade is set from it. The season record starts again.
- **Where it is kept:** `exmortis\giga\` on your PC, with one copy in Steam Cloud.

After each game the WINNER / LOSER plate counts your VP change, with the game's own RANK UP and RANK DOWN animation.

Where your Ranked values show

- The badge next to your name in a Ranked fight is your Ranked grade.

- **Player Data** (Network menu and main menu) shows your Ranked VP, grade and season record in place of the original Ranked rows. Another player's page also shows `YOU VS THIS PLAYER`, your wins and losses against them this season.
- In Player Match rooms, TOP VP and the player cards show Ranked values too.



9. Training

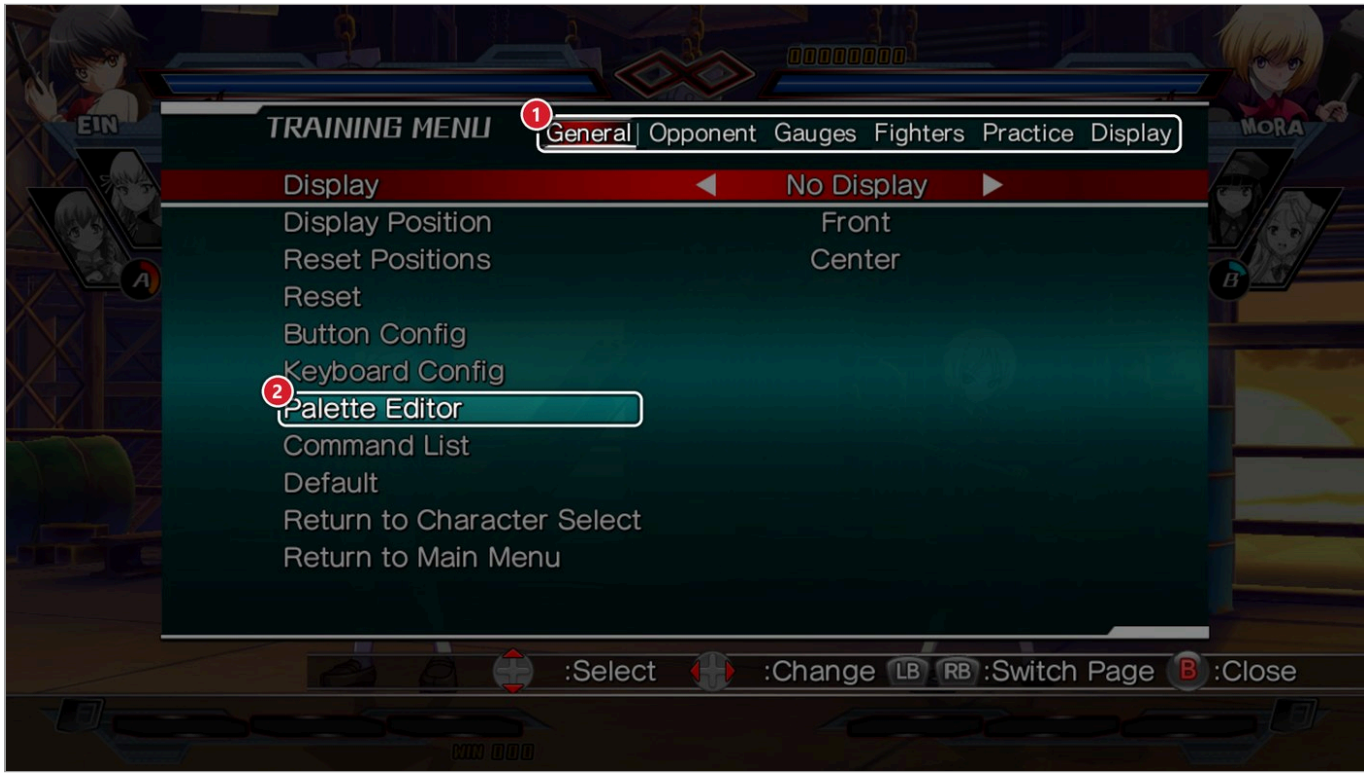
The Training tools work in three places:

- **Training** from the main menu.
- **A room's Training:** the Training Mode row on a room page while you wait.
- **Ranked Training:** Confirm on your own row in the Ranked lobby.

They do not work in Versus, Arcade or online fights. Practice rooms have their own small pause (section 6).

The Training menu

Press Start in Training. The menu has six tabs. General, Opponent, Gauges and Fighters are the game's own pages. **Practice** and **Display** are new. Change tab with Previous Tab / Next Tab (L1 / R1, or K / L).



1. The six tabs. Practice and Display are added by ExMortis.
2. Palette Editor opens the colour editor (section 11).

Also new on the General tab:

- **Meter Gain.** With the game's Display row on Damage Only or Display All, the Damage panel gets a third row: the power the attacker gained in the combo.
- **Input display.** The game's input display is redrawn as a narrow strip at the edge of the screen, with a frame count for each input.
- **Reset to a corner.** Hold Left or Right and press Reset: both fighters start in that corner, for this Reset only.

Practice tab



Row	Values	What it does
Pause	Off, On	Keeps the fight stopped after the menu closes
Frame Step	Confirm	Advances one frame, then stays paused
Speed	100%, 75%, 50%, 25%	Slow motion, in the fight only
Input Delay	0 to 9	Adds delay to your inputs, to practise for online play
Playback Timing	Off, Wake-up, After Block, After Hit, Air Tech	The dummy acts on its own settings until that moment, then plays its recording once
Slot	1 to 4	The state slot that Save State and Load State use. Shows Empty, Saved or Can't load.
Save State	Confirm	Copies the whole fight into the slot
Load State	Confirm	Puts the fight back as it was saved
Auto Load	Off, After Combo	Loads the slot about a second after each combo ends

Frame Step, Save State and Load State run when the menu closes, so Confirm on them closes the menu too. Each row has a help line at the bottom, and a result such as `State saved.` shows there.

How long a saved state lasts. A state can be loaded only in the fight it was saved in, and only until the next Reset. A Reset, Character Selection, a change on the Fighters tab and a change of most Training settings turn the slot to `Can't load`. Changing the dummy's guard and tech settings keeps it. Nothing is written to disk.

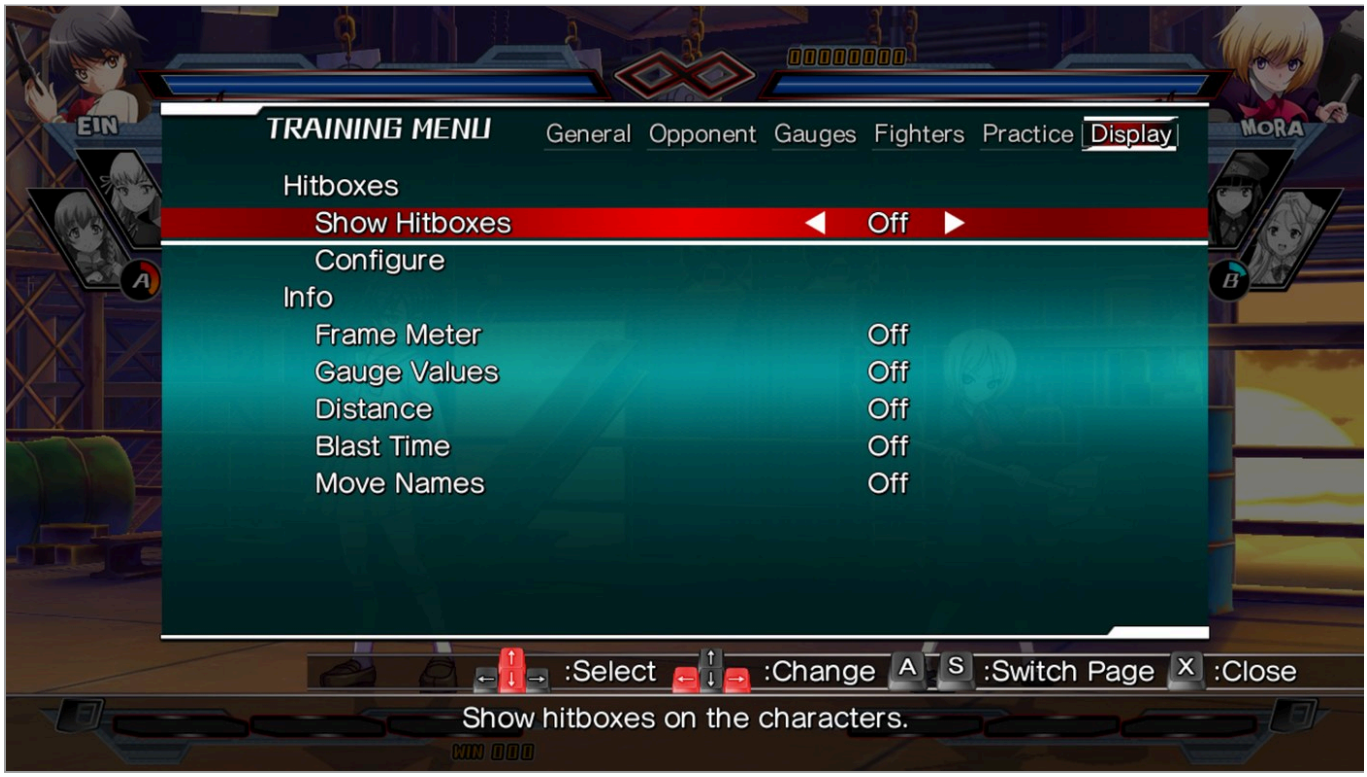
None of the Practice rows is saved: they go back to their defaults when you close the game.

Training keys

Key	Action
F5	Save State
F9	Load State
F6	Pause on / off
F7	Frame Step
F8 (hold)	Rewind: runs the fight backward through the last 10 seconds. Release to play on from there. While paused, a tap goes one frame back.

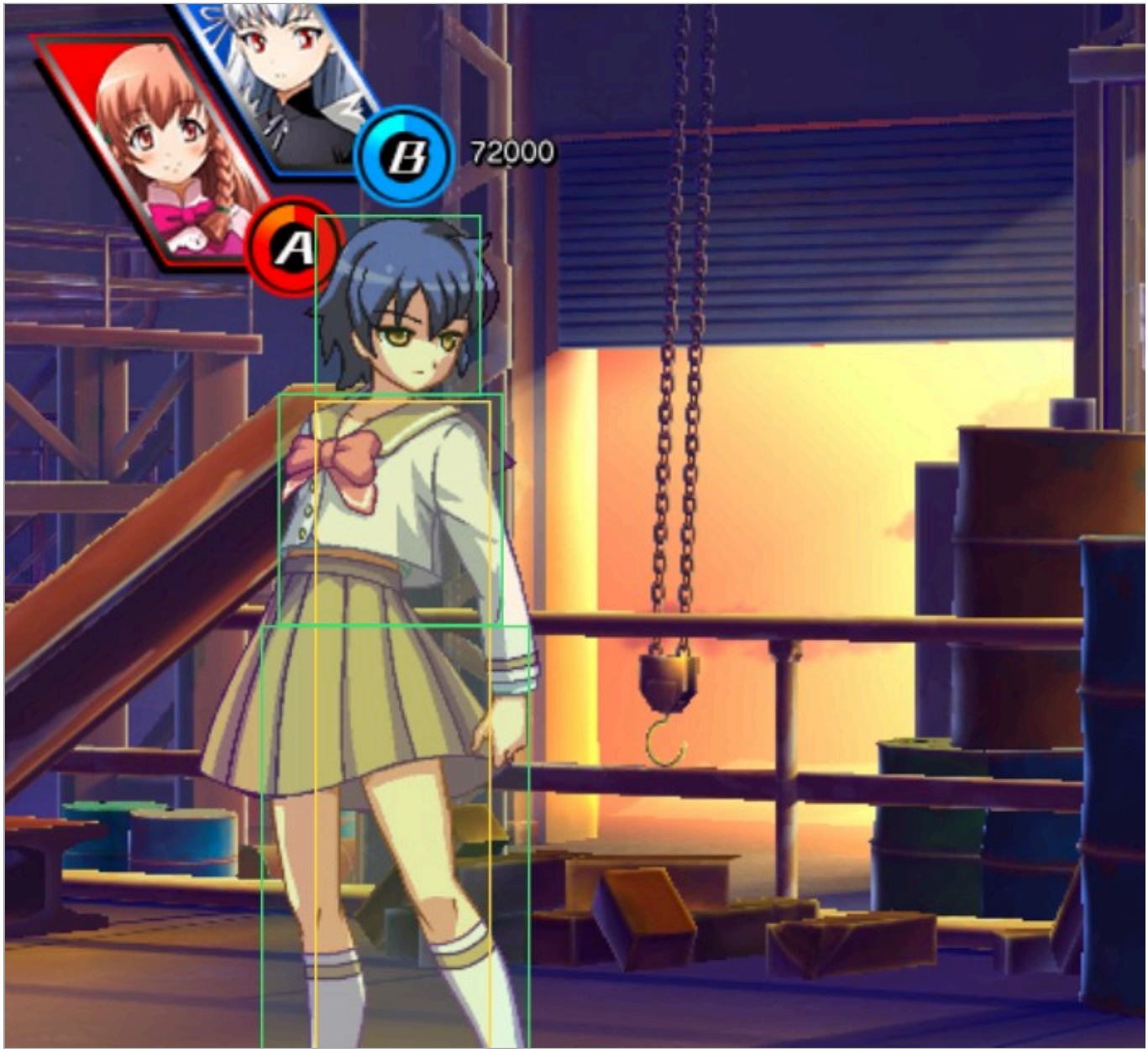
The keys work in the fight with the menu closed. They show no message in the fight: the result is on the Practice tab's help line the next time you open the menu. You can change them in Keyboard Config > Battle Controls, where a **Show Hitboxes** row waits for a key of your choice.

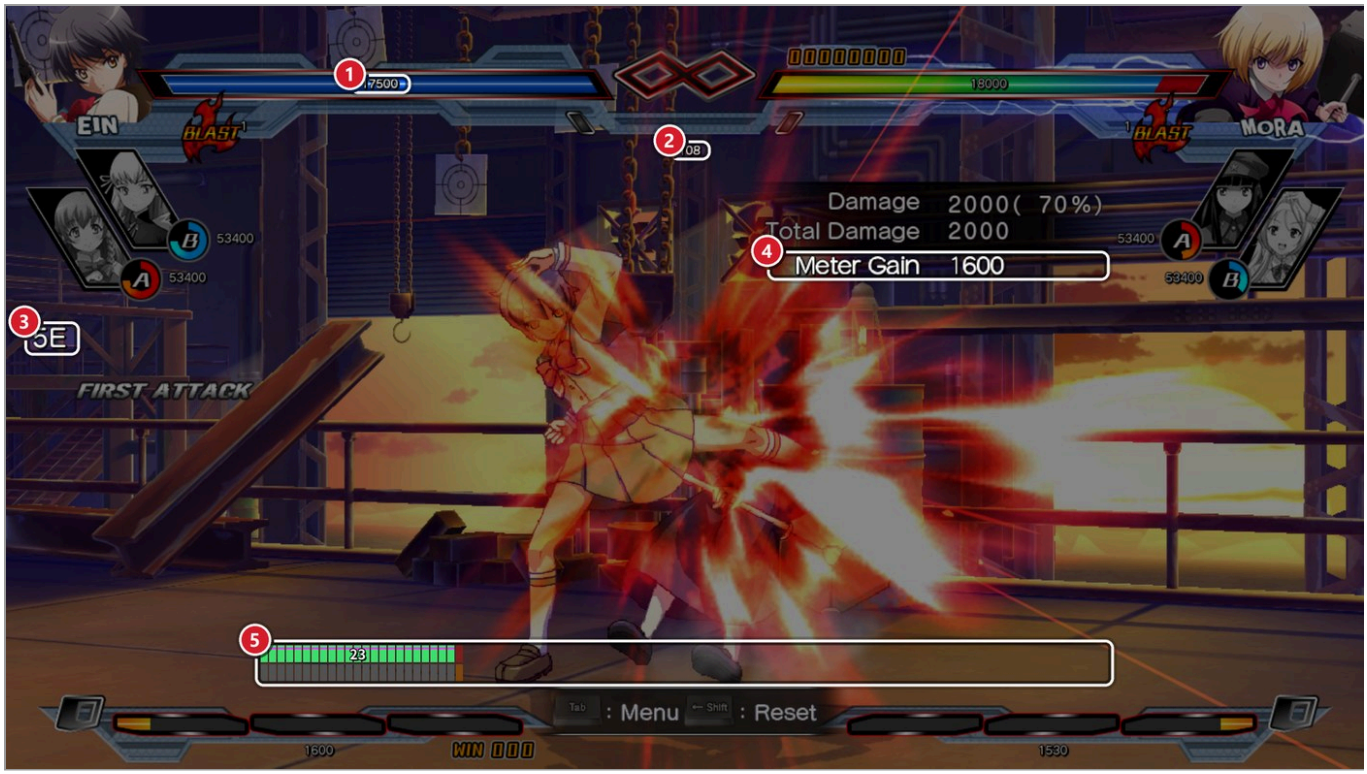
Display tab



Row	What it shows
Show Hitboxes	Boxes on both fighters: red attack boxes, green and blue hurt boxes, yellow push boxes, purple special ranges
Configure	A list: show or hide each kind of box, choose the players (both, 1P only, 2P only), set the fill (Off, Low, Medium, High)
Frame Meter	One cell per frame for each fighter (below)
Gauge Values	Health, power, partner gauges and Blast as numbers on their bars
Distance	The distance between the fighters, under the timer
Blast Time	The time left of a running Blast
Move Names	The name of each fighter's current move

All switches are Off when the game starts and last until you close it. The Configure choices are remembered. Boxes and numbers hide while a pause menu is open.





1. Gauge Values: the numbers on the health bars, partner gauges and power meters.
2. Distance.
3. Move Names: the move being done.
4. Meter Gain, the new row in the game's Damage panel.
5. The Frame Meter.

The same switches apply in replays (the replay pause menu has its own Display list) and in Practice room fights.

Frame Meter

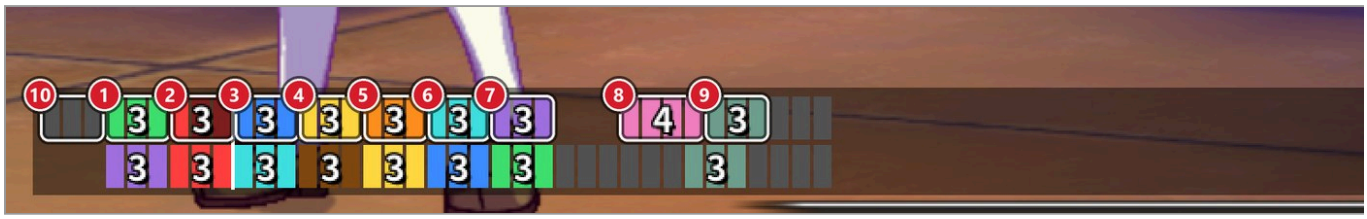
The Frame Meter shows, frame by frame, what each fighter is doing and who can act first. It sits at the bottom of the screen, above the gauges. The top row is 1P, the bottom row is 2P. Each cell is one frame, and time runs from left to right.

Turn it on with Display > Frame Meter. In a replay: pause menu > Display > Frame Meter. In a Practice room: the pause window's Display list.



1. The Frame Meter. The close-ups below show it enlarged.

What the colours mean



No.	Colour of a cell	Meaning
1	Green	Startup: the move has begun, the attack is not out yet
2	Red	Active: the attack can hit on this frame
3	Blue	Recovery: the attack is over, the fighter cannot act yet
4	Yellow	Guard stun: blocking, cannot act
5	Orange	Hit stun: being hit, cannot act
6	Cyan	The move was parried or clashed, and any normal attack can start
7	Purple	Busy with something else, such as a back dash or getting up
8	Pink	Jump startup
9	Grey-green	Landing
10	Dark grey	Free: the fighter can act
	Any colour at half brightness	That fighter is frozen on this frame (hit-stop on one side, a super flash). The last red cell in the picture is one.

Two small marks sit between cells:

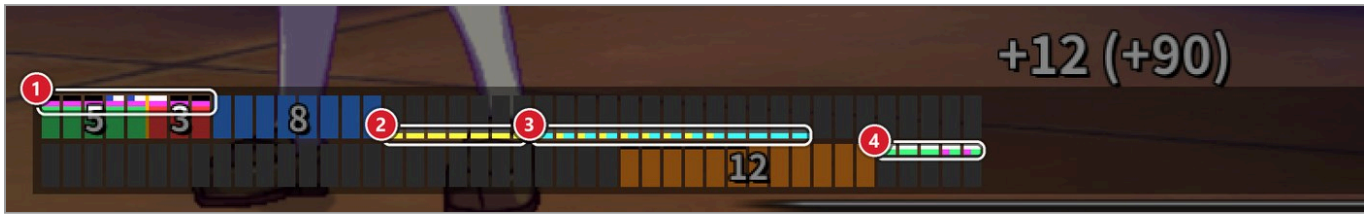
- A **white mark** means hit-stop that both fighters shared was left out there, so the bar stays short.
- An **amber mark** before a cell means this move cancelled the one before it.

The thin strips inside a cell

A cell can carry up to three thin strips. They add to the colour and tell you more about that frame.

Strip	Colour	Meaning
Top	Black	No ordinary attack can hit this fighter (invulnerable to strikes)
Top	Dark blue	Projectiles pass through this fighter
Top	White	An ordinary throw cannot take this fighter
Middle	Magenta	A hit now would be a counter hit
Middle	Green	A hit now would be parried
Bottom	Yellow	A projectile of this side is out and can hit
Bottom	Cyan	This side's partner is out and can hit

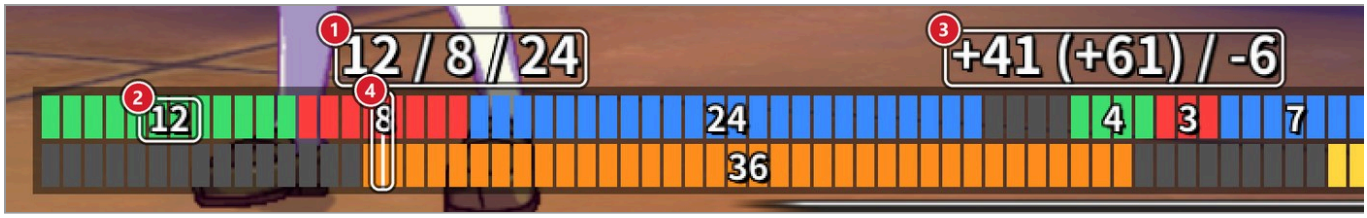
So the top strip is about what can hit this fighter, the middle one about what a hit would do, and the bottom one about what else this side has on screen.



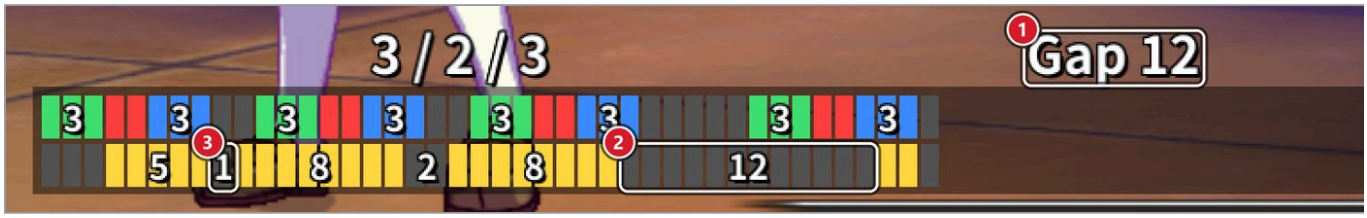
1. Top and middle strips on 1P's move: magenta (a hit would be a counter hit), white (throws miss) and black (strikes miss).
2. A yellow bottom strip: 1P's projectile is out.
3. A cyan bottom strip: 1P's partner is out.
4. Strips on 2P's cells after the hit stun: green (would be parried), white and magenta.

The numbers

- **A digit inside a run of cells** is the length of that run in frames.
- **The move's numbers**, for example `5 / 4 / 9`, are the startup, active and recovery frames of the last finished move. A whiff, a hit and a block read the same.
- **The big number over the bar** is the frame advantage after the last exchange. Plus means you can act first, minus means your opponent can. In Training it is counted for the side you play, in a replay for 1P.
  - `+18 (+114)`: the second number is counted from when you had already recovered, for example after a projectile hit late.
  - `-3 / +1`: the second number is counted from landing, when one fighter was in the air.
- `Gap 3` means your blockstring has a hole: the opponent was free for 3 frames between two blocked attacks. A true blockstring shows no gap.



1. The last move's startup, active and recovery frames.
2. A digit inside a run: this startup run is 12 frames long.
3. The frame advantage.
4. The white mark: shared hit-stop was left out here.

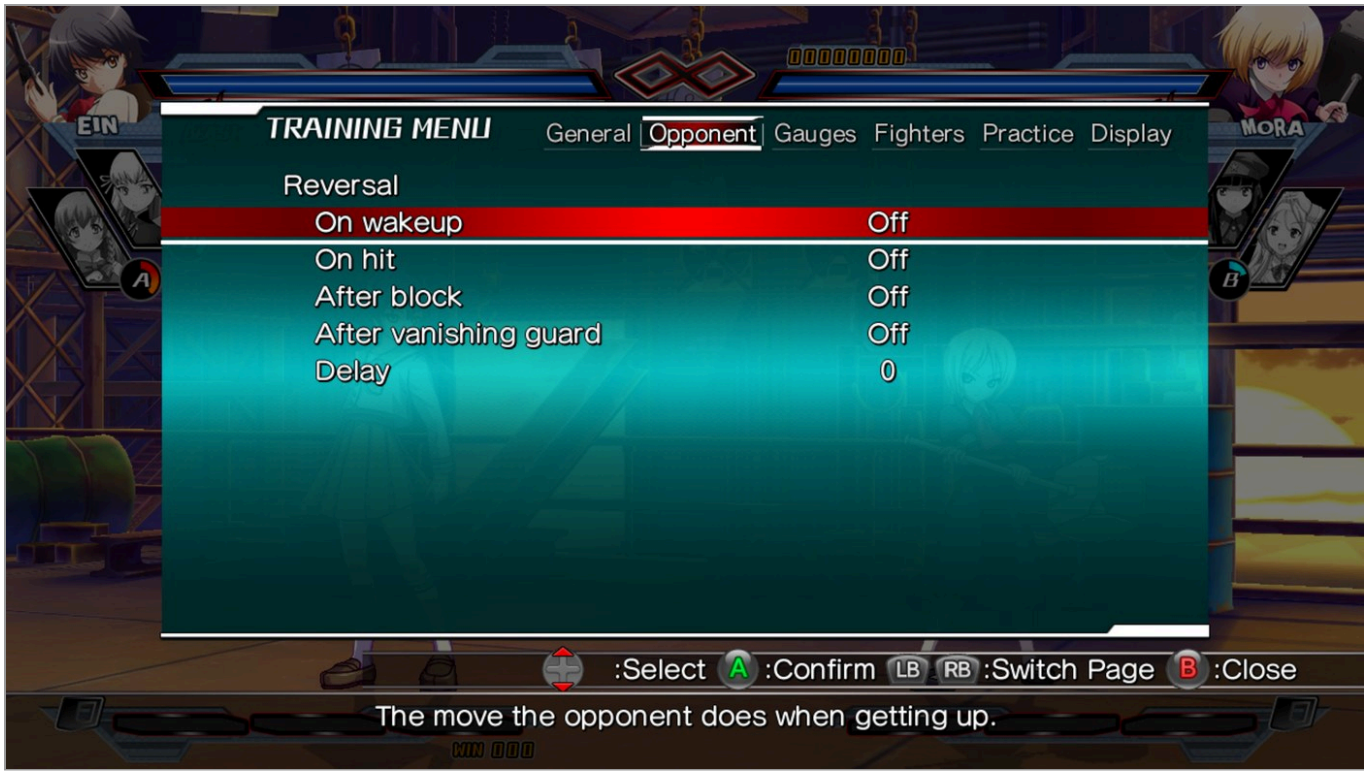


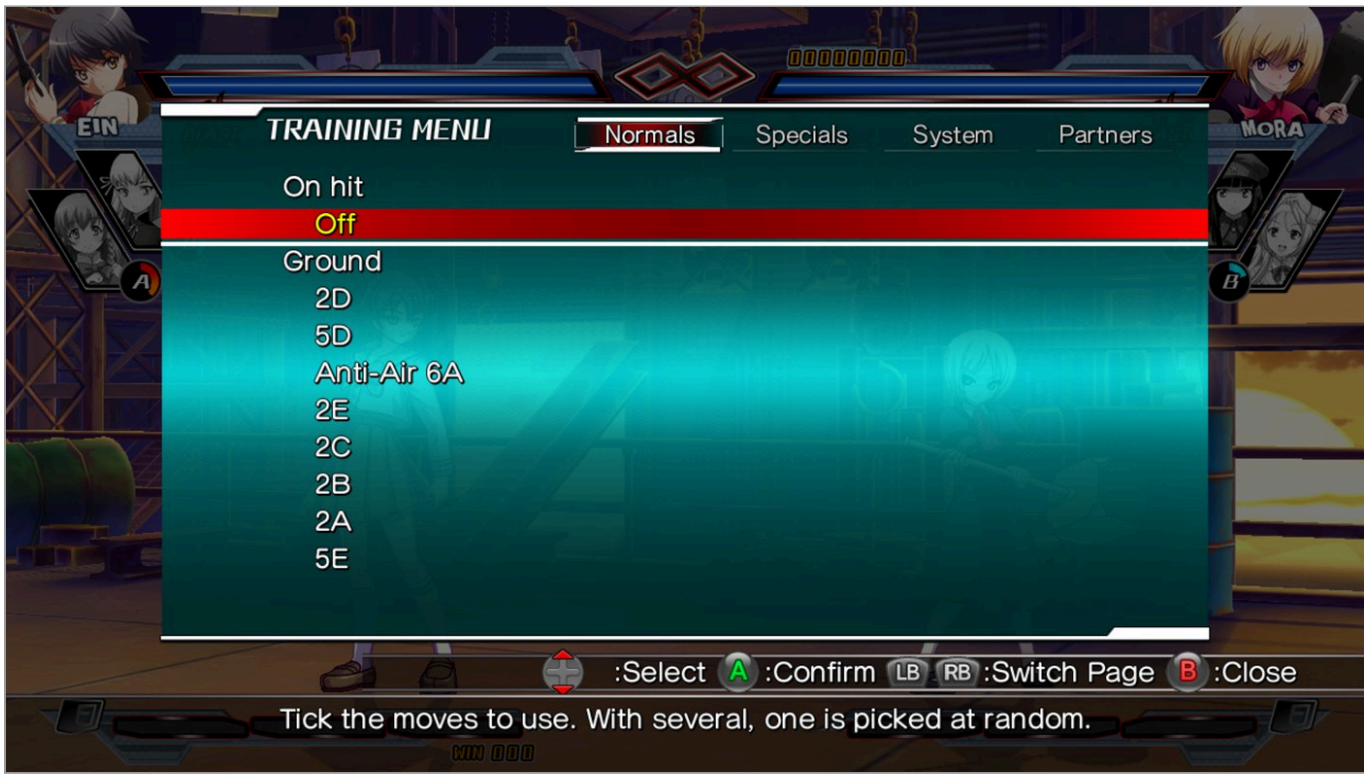
1. `Gap 12`: the last hole was 12 frames long.
2. That hole in 2P's row: 12 free cells between two guard stuns.
3. A hole of one frame.

Reversals

Make the dummy do a move at a chosen moment.

1. Open the **Opponent** tab and press Confirm on **Reversal**.
2. Pick the moment: **On wakeup**, **On hit**, **After block** or **After vanishing guard**. Press Confirm.
3. Tick the moves with Confirm. The list has tabs (Normals, Specials, System, Partners), and only moves that fit the moment are listed.
4. Press Back to close the list.





- With several moves ticked, the dummy picks one at random each time.
- `Off` at the top of the list clears that moment.
- **Delay** (0 to 10 frames) makes the dummy wait before the move.
- The picks belong to the fight. A new fight, another dummy or other partners clear them.
- A Reversal is quiet while a recording records or plays. If Playback Timing and a Reversal fall on the same moment, the recording wins.

Dummy recordings

The game's Start Recording and Playback rows work as before, with four fixes:

- A recording can be 99 seconds long (it was 30).
- Each of the four slots keeps its own length.
- Recordings are kept per dummy lineup (fighter and both partners) in `exmortis\training`.
- They come back the next time you start the game.

10. Replays

Player Match and Ranked games can be saved as replays and watched in the game's Replay Theater.

Saving replays

There are two ways.

- **Save every match:** Options > Game > **Save Replays** on. It is off by default. A change counts from the next game.
- **Save one match:** with Save Replays off, the online result screen shows a **Save Replay** line under Input Delay. Press Start (Tab or Enter on the keyboard). The line then reads `Replay saved.` The chance ends when the next match starts.

What is saved, and what is not:

- A match that broke off (a disconnect or a desync) is saved too, when Save Replays is on. It is marked DISCONNECTED or OUT OF SYNC and plays up to where it stopped.
- A match longer than about 15 minutes is saved in part (PARTIAL).
- Practice room fights and matches you only watched are never saved.

The files are in `exmortis\replays\Player Match` and `exmortis\replays\Ranked`. A file name holds the date, both players, their fighters and the score.

The Replay Theater

Network > Replay Theater. It opens on the PLAYER MATCH tab. RANKED MATCH, and the game's own BATTLE LOG and LEGACY, show only while they hold replays.



1. The tabs, each with its count. Change tab with Previous Tab / Next Tab.
2. A star marks a replay you kept.
3. The card of the selected replay: fighters, partners, stage, match length and ping.

The RESULT column shows WIN, LOSE or DRAW from your side. A match you did not play in shows no word.

Key	Pad	Keyboard 1	What it does
Confirm	Cross / A	`U`	Open the replay's Details
Left / Right	D-pad	`A` / `D`	Previous / next page
Start	Options / Start	Tab	Keep, or stop keeping
Reset	Share / Back	Left Shift	Delete (asks first)
1P Player Data	Square / X	`J`	Copy the replay file to the clipboard
Prev. / Next Category	L2 / R2	`E` / `Q`	Sort by Date, Opponent or My Character
Back	Circle / B	`I`	Leave

How many are kept. Your newest 100 replays per tab stay. Older ones are removed as new ones arrive. **Keep** stores a replay for good, up to 100 more per tab. At 100 kept, the next Keep shows `No more replays can be kept. Remove Keep from one first.`

Details



Details shows both fighters, names, partners, stage, custom or original colours, the result, the round winners, match length and ping. The buttons are **PLAY**, **KEEP** (or UNKEEP), **DELETE** and **BACK TO LIST**. Left and Right pick a button, Confirm presses it.

Sharing replays

- **Copy:** on the list, press 1P Player Data. The file is on the clipboard: paste it in a chat or a folder.
- **Add:** drag a `.nbreplay` file onto the game window. It lands in Keep. A drop is refused during an online match, when 100 are kept, or when a file of that name is already there. `File added.` shows only while the Theater is open.

Watching a replay

A replay answers the menu keys of the player who started it in the Theater.

Key	Pad	Keyboard 1	What it does
Confirm	Cross / A	`U`	Play / pause
Left / Right (tap)	D-pad	`A` / `D`	Pause and step one frame back or forward
Left / Right (hold, while playing)	D-pad	`A` / `D`	Scrub backward or forward at 2x, then 4x, then 8x
Previous Tab	L1	`K`	Go to the start of this round (in a round's first two seconds: the round before)
Next Tab	R1	`L`	Go to the next round
Speed Up	R3	`H`	1x, 2x, 4x, 8x, then 1x again
Speed Down	not set	`G`	One step slower, down to 1x
Take Control	L3	`T`	Play one of the fighters yourself (below)
Start	Options / Start	Tab	The pause menu
Mouse			Click or drag the position bar to jump

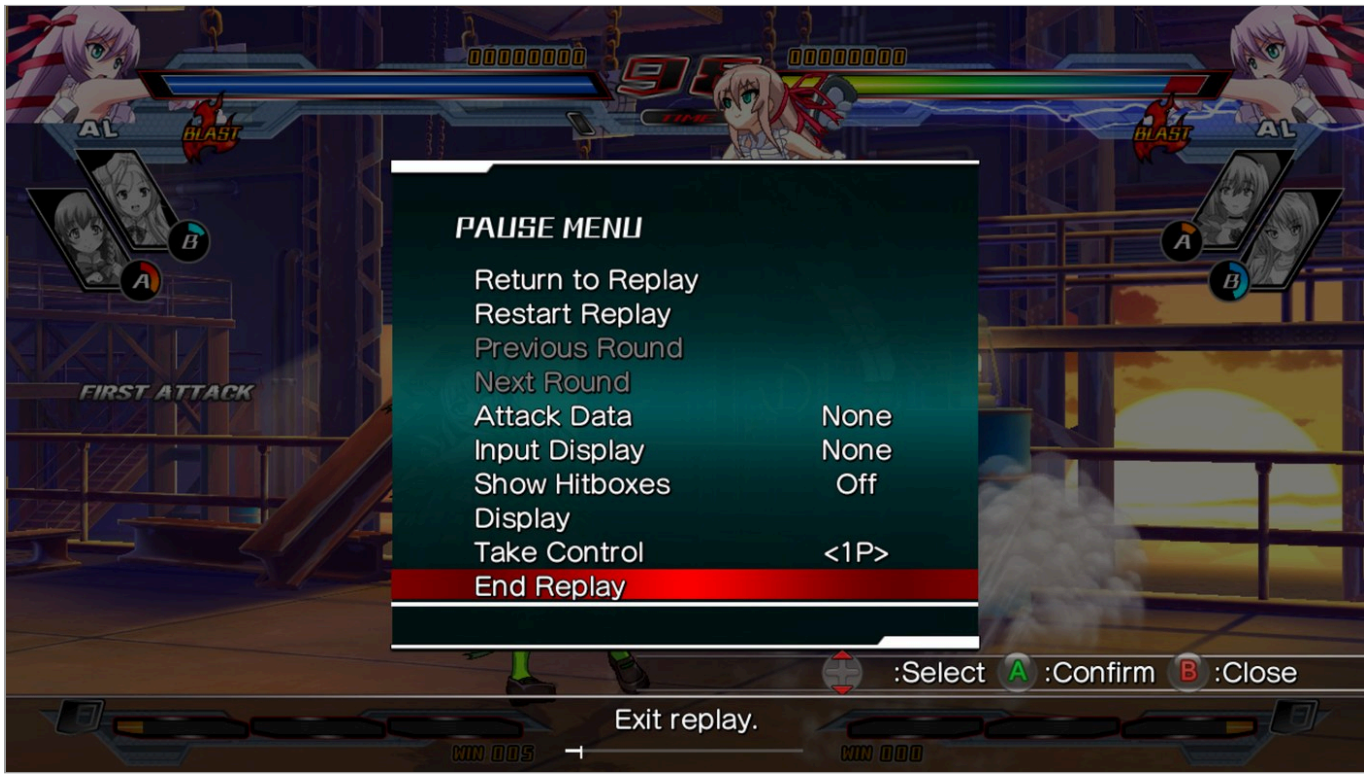


1. The position bar and the state (`Paused`, `2x`, `-4x` and so on). It appears when you press a replay key or move the mouse, and fades after two seconds. Click or drag it to jump.

Good to know:

- A frame step and the end of a scrub set the speed back to 1x.
- Sound is muted at 4x and above and in rewind.
- A jump far ahead runs behind a black screen for a moment.
- Replays and watches show the custom colours the players wore.

The replay pause menu



Row	What it does
Return to Replay	Close the menu
Restart Replay	Start again from the first frame
Previous Round / Next Round	Jump between rounds. Grey when there is no such round.
Attack Data	The game's damage panel: None, 1P, 2P or both
Input Display	The input strip: None, 1P, 2P or both
Show Hitboxes	On / Off
Display	A list: Frame Meter, Gauge Values, Distance, Blast Time, Move Names
Take Control	Choose 1P or 2P with Left / Right, then Confirm
End Replay	Back to the Theater

Show Hitboxes and the Display list start from what you set in Training. Attack Data and Input Display start at None in every replay. What you change lasts until the replay ends.

When the replay reaches its end, it holds on the last frame and a menu opens by itself: Restart Replay, Previous Round, the display rows and End Replay.

Take Control

Take Control lets you play one fighter from the frame on screen. The other fighter keeps doing what the recording says.

1. Press the Take Control key and choose 1P or 2P. Or use the pause menu's Take Control row.

2. A one-second countdown runs. Then your **player 1 battle controls** drive that fighter.
3. To try again from the same frame, press Take Control again (a new countdown) or Load State, F9 (at once).
4. To go back to watching, open the pause menu and choose **Return to Replay**.

Nothing you play in control is saved. If your Take Control key is also one of your battle keys, it does nothing while you are in control: use the pause menu's Retry.

Replays from other versions, and damaged files

- A replay from **another mod build** still plays, but it is not checked. It can play differently from the real match, and nothing on screen says so.
- A replay in **another file format** is listed as DAMAGED with `The version information differs. Unable to use this replay.` It cannot play. Keep, Delete and Copy still work.
- A **cut-off or broken file** reads DAMAGED, or gives `The replay file is damaged.` when you press Play.
- `Replay stopped before completion.` means the replay did not match its own recording part way through.

The game's own replays

BATTLE LOG and LEGACY (the game's My List) are the game's own replay lists. They are drawn in the same style, and most of them play with the same keys, bar, pause menu and Take Control. The mod never changes these files. Battle Log does not grow any more, because the game's own Ranked Match is replaced.

11. Colours

Every fighter can have up to 64 colours of your own. They are numbered 22 to 85, after the game's 21.

Picking a colour

On any Character Selection, step through the colours with Left and Right as usual. After the stock colours come yours, shown as `Color 22`, `Color 23` and so on. Confirm picks it.

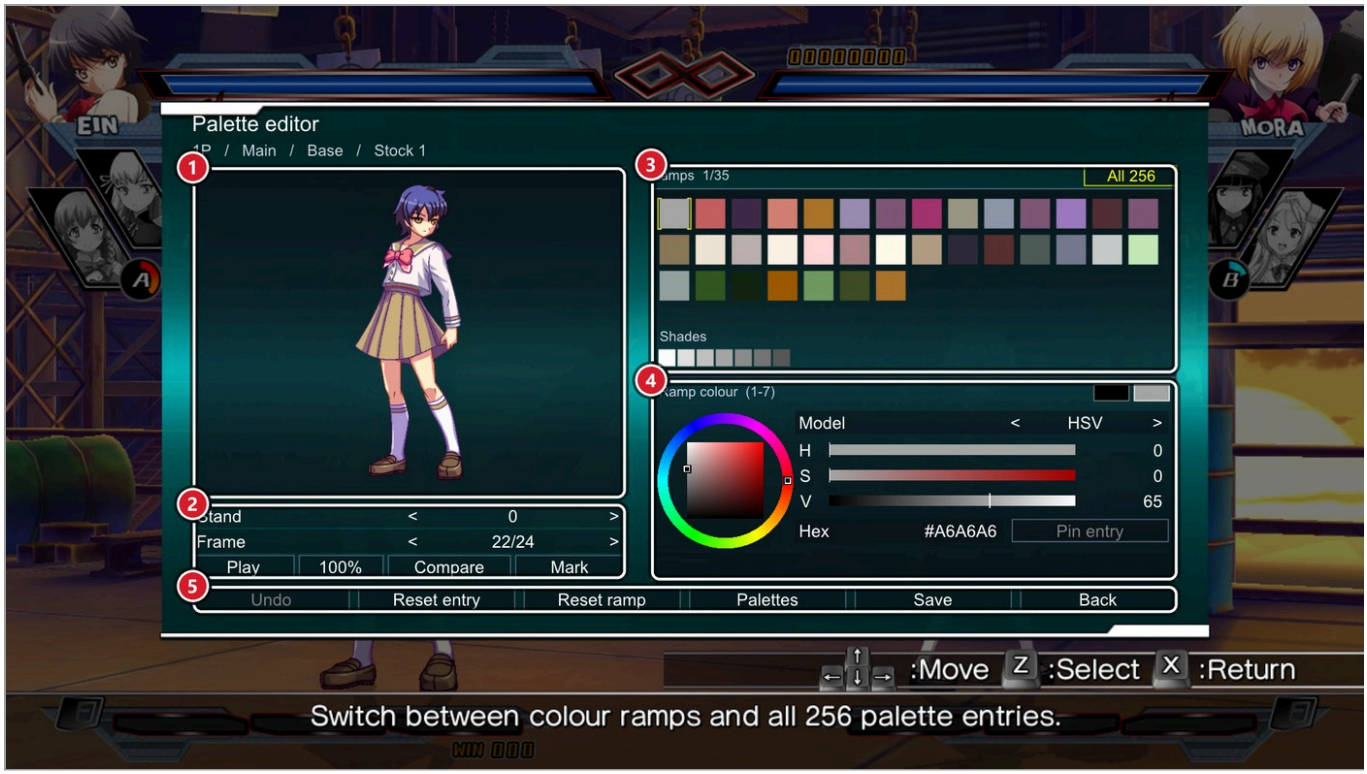
- Only the number is shown, not the colour's name.
- A custom colour is offered only if the stock colour it was made from is unlocked.
- Your pick is remembered for that side and fighter until you close the game.
- If both players pick the same custom colour for the same fighter, 2P gets the stock colour.
- Custom colours change the fighter only. Partners keep their own colours.

Colours online

- In Player Match, each player picks from their own colours, and each PC shows both players' colours. Only the colours are sent. No file is written on the other PC.
- A colour you make or add while you are in a room is offered two Character Selection visits later, on both PCs at the same visit. Nothing changes during a fight.
- Watchers and replays show the colours the players wore.
- To hide one player's custom colours on your screen, use Options > Game > **Player Colours** (section 13). Your own fighter always keeps your colour.

The palette editor

Open it from Training: pause menu > General > **Palette Editor**. It also works in a room's Training. The editor covers the paused fight, and every change shows on your fighter at once. Its texts are in English only.



1. The preview of your fighter.
2. Animation, Frame, Play and Zoom choose what the preview shows. Compare puts the original next to your version. Mark flashes the colour you are editing.
3. The ramps: each row is one group of shades, such as hair or a skirt. `All 256` shows every entry.
4. The colour editor: HSV or RGB sliders, a colour wheel, a Hex field and Pin.
5. Undo, Reset entry, Reset ramp, Palettes, Save and Back.

To change a colour:

1. Move to a ramp and press Confirm. Its shades open.
2. Move to a shade and press Confirm. You are on the sliders.
3. Change H, S and V with Left and Right. Hold Start for steps ten times larger. A tap of Start swaps HSV and RGB.
4. Press Back to step out one level.

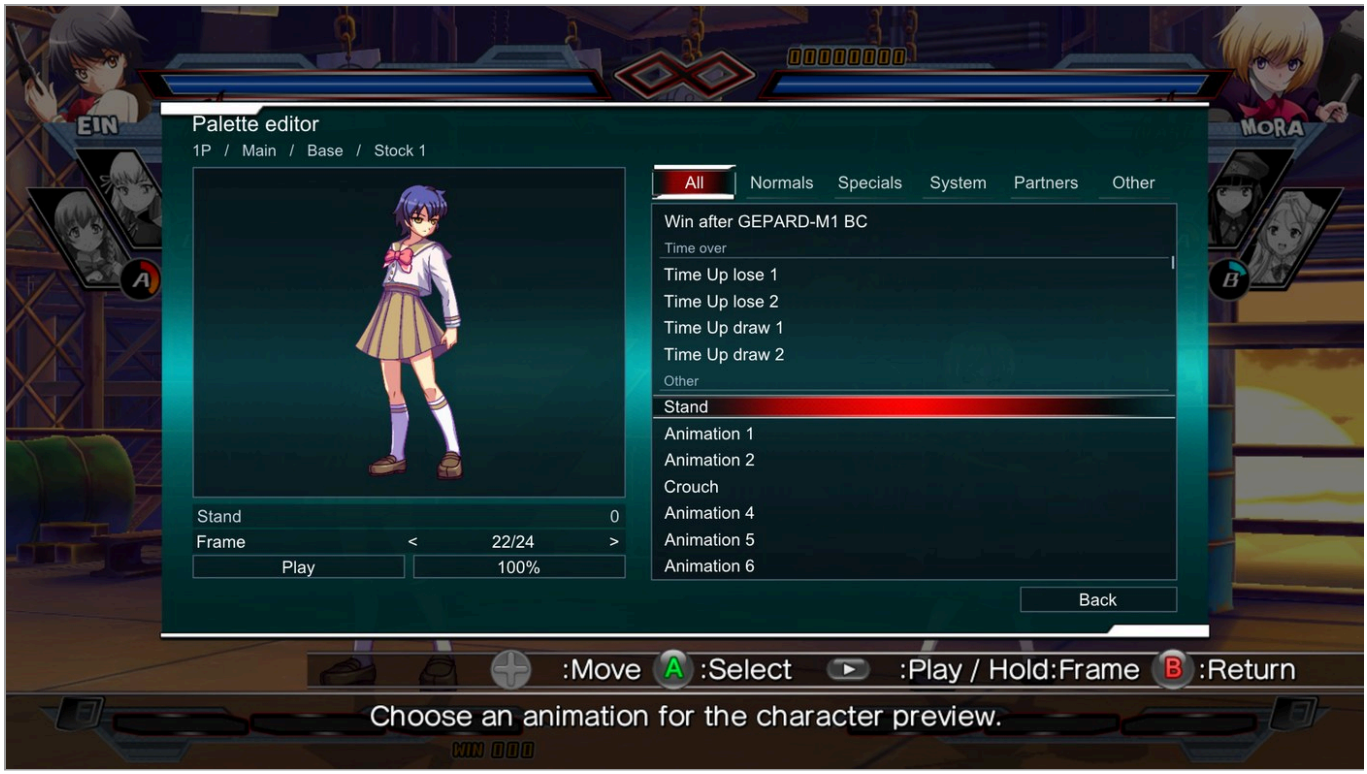
Changing a ramp's colour moves all its shades together. A **pinned** entry keeps its colour. Undo goes back up to 16 steps. Reset entry and Reset ramp ask first.

With a mouse: click cells and buttons, drag the sliders and the wheel. On the preview, the mouse wheel zooms, a drag moves the sprite, and a click picks the colour under the pointer.

Hex: press Confirm on Hex and type six digits. Enter accepts, Escape cancels.

Checking a colour on other moves

Press Confirm on the **Animation** row. A list of the fighter's moves opens, by name, with tabs: All, Normals, Specials, System, Partners, Other. Round intros and win poses are in it too.

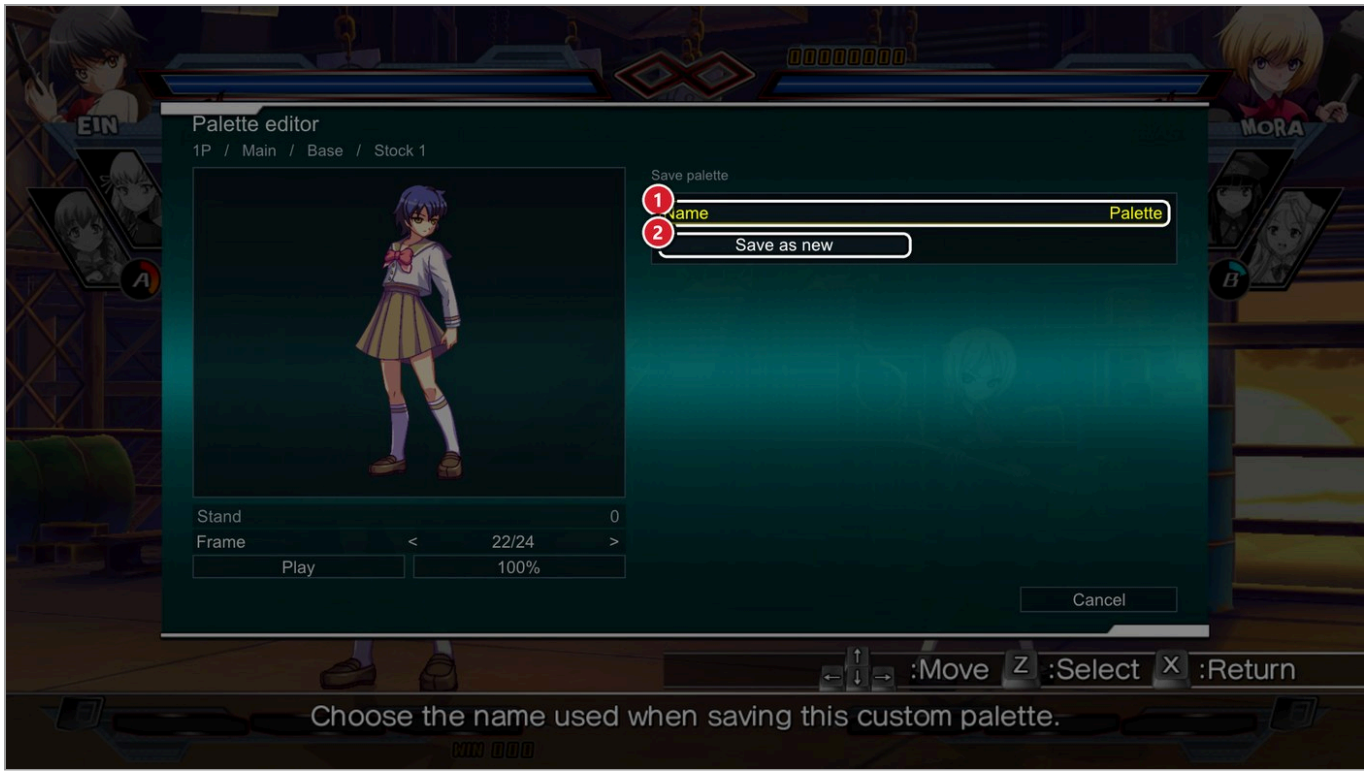


- Up and Down move through the list, and the preview follows.
- Left and Right change the tab.
- Confirm keeps the move. Back puts the old one back.
- A tap of Start plays or pauses. Start held with Left or Right steps one frame.

Saving a colour

1. Press **Save** in the footer.
2. Press Confirm on **Name** and type a name (Enter accepts). On a pad, Up and Down change a letter.
3. Choose **Save as new**. The colour takes the lowest free number from 22, and your fighter wears it at once.

With a custom colour loaded, **Replace custom N** saves over it instead.

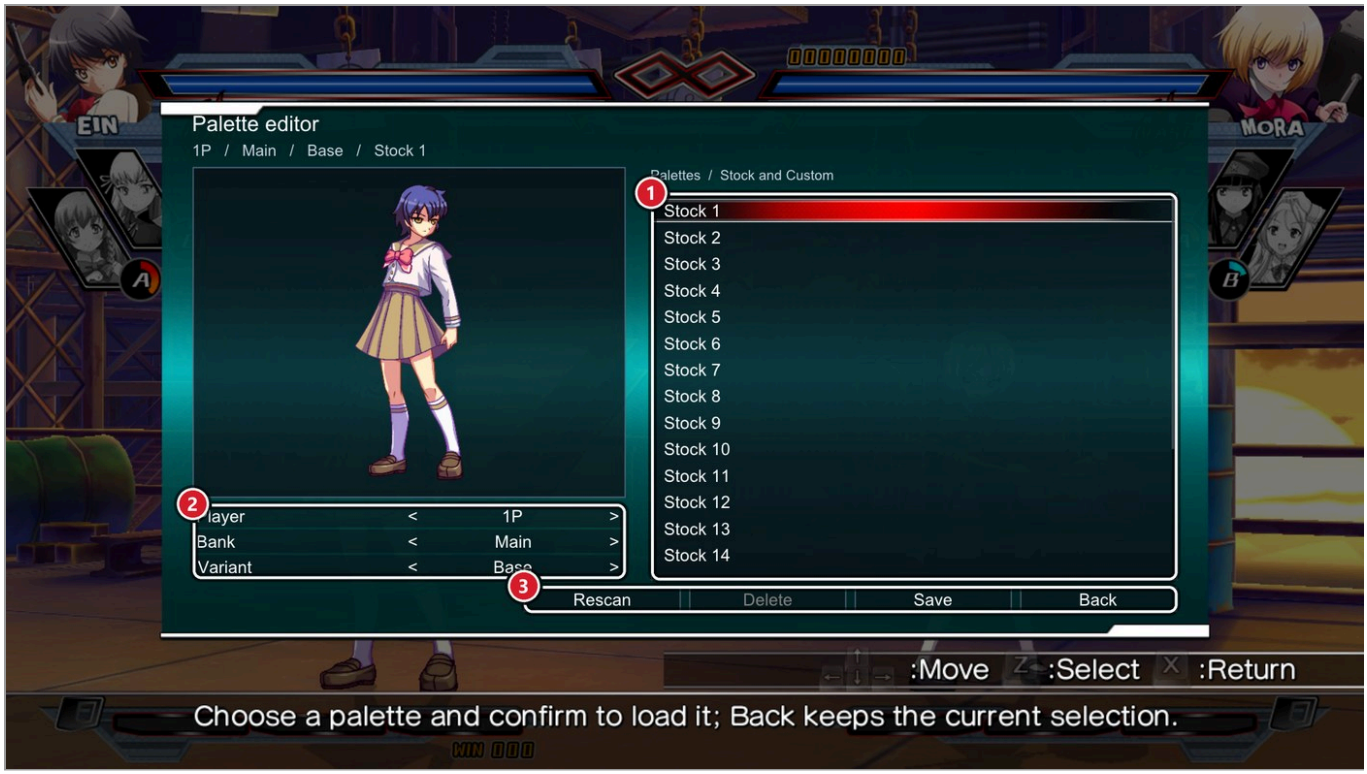


1. Name: the colour's name.
2. Save as new.

If you leave with unsaved changes, the editor asks: **Keep editing** or **Discard and continue**. Discard goes back to your last Save, or to how the fighter looked when you opened the editor if you saved nothing.

The Palettes page

Press **Palettes** in the footer. The list shows everything that can be loaded for the fighter: Stock 1 to 21, then your Custom colours with their names.



1. The list of colours that can be loaded.
2. Player, Bank and Variant.
3. Rescan, Delete, Save and Back.

- Confirm on a row loads it.
- **Player**, **Bank** and **Variant** under the preview choose 1P or 2P, the colour bank (Main, Sub1, Sub2) and the costume.
- **Delete** moves the loaded custom colour to the `deleted` folder, after a question.
- **Rescan** reads the folder again.
- Start on a Custom row copies its file to the clipboard (`File copied.`).

Colour files

- A colour is one `.nbpal` file in `exmortis\palettes`.
- **To share one:** copy it with Start on the Palettes page and paste it in a chat or a folder. Or send the file from the folder.
- **To add one:** drag the `.nbpal` file onto the game window, or put it in `exmortis\palettes` and press Rescan. A drop is refused during an online match.
- **To get a deleted one back:** move it from `exmortis\palettes\deleted` back to `exmortis\palettes` and press Rescan.
- A fighter holds 64 colours. A 65th file is left out.

12. Controls

Keyboard Config

Open it from Options (KEYBOARDCONFIG), or from the pause menu in Training and Versus. It has two lists: **Battle Controls** and **Menu Controls**.



1. Battle Controls: the keys you fight with, and the shortcut keys. Menu Controls: the keys for menus and replays.

Menus never read the battle keys, so changing your battle layout does not move Confirm or Back.

To change a key:

1. Move to a row and press Confirm.
2. Press the new key within 5 seconds. Escape cancels.
3. When you are done, press **Start**, or Confirm on **Confirm Settings**, to save. **Back undoes everything** you changed on this visit.

Situation	What happens
The key is already yours, or it is a shortcut key	The two rows swap.
The key is the other player's battle key	The key moves to you and their row is cleared.
Left or Right on a row	The row is cleared. Confirm, Back, the tab keys and the menu directions cannot be cleared.
Escape or a Start key	`That key is reserved. Choose another key.`
Default Settings	Battle Controls: your battle keys and the shortcut keys. Menu Controls: your menu keys. The other player's keys stay.
During an online match	The page is grey: `Not available online.`

The keyboard's Start key (Tab, Enter) and battle Back key (Left Shift, Right Shift) are fixed.

Default keys

	Keyboard 1	Keyboard 2	Pad
Move	W A S D	Arrow keys	D-pad and left stick
Battle A B C D E	J O I L U	Num 4 9 8 6 7	Button Config
Battle AD, BD, ABC	K, E, Q	Num 5, Num 2, Num 1	Button Config
Battle Back, Start	Left Shift, Tab	Right Shift, Enter	Share / Back, Options / Start
Menu Confirm, Back	U, I	Num 7, Num 8	Cross / A, Circle / B
Menu Previous Tab, Next Tab	K, L	Num 5, Num 6	L1, R1
Menu Player Data 1P, 2P	J, O	Num 4, Num 9	Square / X, Triangle / Y
Menu Reset	Left Shift	Right Shift	Share / Back
Menu Prev. / Next Category	E, Q	Num 2, Num 1	L2, R2
Replay: Take Control, Speed Up, Speed Down	T, H, G	Num 0, Num 3, Num .	L3, R3, not set

Shortcut keys, shared by both keyboards:

Key	Action	Where it works
F5	Save State	Training
F9	Load State	Training (Retry while you have taken control in a replay)
F6	Pause	Training
F7	Frame Step	Training
F8 (hold)	Rewind	Training
`-` and `=`	Input Delay down / up	Room page, result screen, your online match
not set	Show Hitboxes	Training, Practice room fights, replays

All of them are rows in Keyboard Config > Battle Controls and can be changed.

Button Config (pads)

Button Config also has two lists.

- **Battle Controls** is the game's own editor. New: **L3 and R3** (stick clicks) can be bound.
- **Menu Controls** is the mod's: Confirm, Back, Previous Tab, Next Tab, Player Data, Reset, the two Category buttons, Take Control, Speed Up, Speed Down.

On Menu Controls, press a button on a row to bind it. **It takes effect at once**, and the row that had the button takes this row's old one. Left or Right clears a row. Start or Confirm Settings saves and leaves. Back undoes what you changed on this visit and leaves.

The d-pad, the stick and Start cannot be changed. Menus always use the menu buttons, whatever your battle layout is. That is what fixes Confirm and Back on PlayStation pads.

Pads and keyboards together

- The device that presses at the title is player 1, as before.
- A pad you switch on later: with the keyboard as player 1, press Confirm on the pad in the Network menu or in a room. It becomes player 1's pad. Not during a match.
- A newly plugged pad is found within a second or two.

13. Settings and files

Options > Game

The mod adds seven rows after the game's four. The list scrolls.

Row	Values	Default	What it does
Input Delay	0 to 5, Auto	Auto	Your input delay online (section 5)
Player Colours	A list	On for everyone	On or Off for each player you have met: show or hide their custom colours
Save Replays	Off, On	Off	Save every online match (section 10)
Watch Matches	Off, On	On	Follow the room's matches while you wait (section 7)
Nicknames	On, Off	On	Show player names in the fight and on the WINNER / LOSER plate. Off also keeps names out of the Discord status.
Delay / Ping	On, Off	On	The delay and ping line in online fights
Discord Status	On, Off	On	Shows what you are playing on Discord. Off: the mod does not talk to Discord at all.



1. The seven rows added by ExMortis.

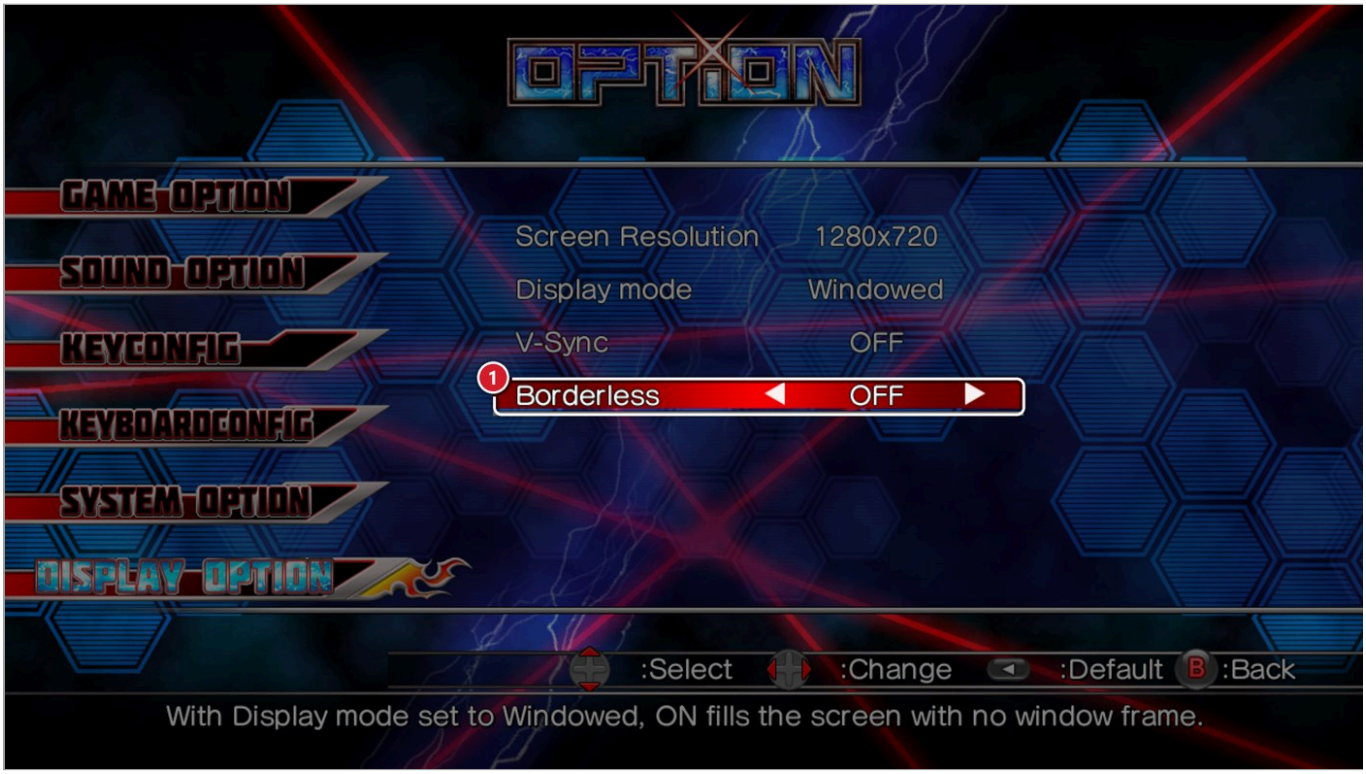
The rows are saved when you leave Options. The page's Default key resets the value rows. It does not change the Player Colours list.



The list holds 64 Off players. To turn off one more, turn another back on first.

Options > Display

Row	Values	Default	What it does
Borderless	Off, On	Off	With Display mode on Windowed, fills the screen with no window frame



1. Borderless, the row ExMortis adds.

If the change cannot be applied, the old setting comes back with a message.

The exmortis folder

File or folder	What it holds
`exmortis.ini`	All settings. The menus write it. You never need to edit it.
`exmortis.ini.bak`	The settings before the last save
`SaveData.dat`	The mod's copy of your save (section 3)
`replays\Player Match\`, `replays\Ranked\`	Replays. Each has a `keep\` folder for the ones you kept.
`palettes\`	Your colours (`.nbpal`). Deleted ones go to `palettes\deleted\`.
`training\`	Dummy recordings
`giga\`	Your Ranked rating
`exmortis.log`, `exmortis.log.1` to `.5`, `bootstrap.log`	Logs: this launch and the five before it
`diagnostics\`	One folder per problem, and `exits.log`

`exmortis.ini` holds the Steam ids of players you have met (for Player Colours and remembered delays). Keep that in mind before you share the file.

If a line in `exmortis.ini` is not understood, the mod skips that line and says so in the log. If the file cannot be read at all, a box at the title says `exmortis.ini could not be read. The game's own controls are in use.` Your next saved change writes a fresh file, and the unreadable one is kept as `exmortis.ini.refused`.

14. Window, display and startup

- **60 frames per second:** menus, selection screens and fights run at an even 60.
- **Startup logos:** press Confirm to skip them and go straight to the title.
- **Resize:** drag any edge or corner of the window. The game keeps its 16:9 picture and adds black bars when the shape differs.
- **Maximize:** the window has the normal maximize button.
- **Remembered:** the window's place, size and maximized state come back at the next launch.
- **Borderless:** Options > Display > Borderless.
- **Alt+Enter** swaps Windowed and Borderless for this run only. The next launch uses the Options row. It is refused in an online match, a watch or a replay, while the game loads, and in Fullscreen.
- **A display that goes away** (a laptop lid closed, a monitor turned off or unplugged, a driver reset): the game keeps running and online matches stay connected. The picture comes back with the display. If the display the game started on is gone for good, the window shows a note asking you to connect it again or restart the game on the other display.
- **A second copy** of the game cannot be started, as in the original.

15. Fixes to the original game

Problem in the original game	With ExMortis
Each frame ended about a millisecond late, so the game ran near 58.8 fps with uneven frames.	An even 60 fps, always on. With V-Sync on, in a window, offline, the pace follows a screen that is close to 60 Hz.
Windows does not count a pad as activity, so the screen could go dark in a long fight.	The PC stays awake during fights, watches and replays. Not in offline Training or in menus.
On some PCs the game stalled for a moment every few seconds: it read its music straight from the drive.	The music is read through the Windows file cache. The stall is gone.
The game asked Windows for new pads every frame.	Empty pad slots are checked in the background.
Training recordings stopped at 30 seconds, were all cut to the length of the last one, vanished at restart and played on the next fighter you picked.	Fixed (section 9).
In Local Versus and Training, Cancel on Character Selection left the screen.	Cancel steps back one pick at a time.
Local Versus results gave one choice for both players.	Each player chooses Rematch, Character Selection or Return.

In a window or in Borderless the game also draws through a newer Windows path that shows each frame with less waiting. There is nothing to set.

16. Discord status

If Discord is running, your status shows what you are doing in the game.

You are	Status line
In an online fight	`Player Match` or `Ranked Match`, then `vs <opponent> (<their fighter>)` and the set score
Watching a match	Both players, their fighters and the set score
Watching a replay	`Watching a Replay`
In the Ranked lobby, in a room, in Training, in an offline match, in menus	A short line for each

- The large picture is your fighter. The small one is the other side's.
- **Invites:** in a Player Match room that is not Ranked, Discord offers **Invite to Play** in a chat and **Ask to Join** on your profile. Taking an invite joins the room, even from a closed game.
- **What Discord receives:** the mode, fighter names, the Steam names and set score of a match or watch, and for a room its player count and join link.
- **Names:** with Options > Game > Nicknames off, no player names are sent.
- **To turn the status off,** set Options > Game > **Discord Status** to Off.

Taking an invite or a join link while you are in a match leaves that match at once. In a Ranked set this ends the set with no rating.

17. Troubleshooting and reporting a problem

Common questions

Problem	Answer
The download is blocked, or `dinput8.dll` disappears	Your browser or antivirus flagged it. See "If the download or the file is blocked" in section 3.
The window title does not show `+ExMortis`	`dinput8.dll` must be next to `Nitroplus Blasterz.exe`, and it must be ours, not another mod's. The game must be the current Steam version.
A fight started by itself in the Network menu	The Performance Test (section 4). Let it finish, or press Start to skip it.
I cannot see my friend's room, a room refuses me, or my friend is not in the Ranked lobby	You do not have the same mod file. Both of you install the same `dinput8.dll`.
The delay number is yellow	Your connection needs more than the delay you have. In a match at 5 it cannot go higher: expect more rollback.
My keys are back to default	See `exmortis.ini` in section 13.
My settings, replays or colours are gone after an update	They are in the `exmortis` folder next to the game. Keep that folder when you replace the DLL.
My unlocks are missing	The mod uses its own copy of your save (section 3).
Alt+Enter does nothing	It is refused during online matches, watches, replays and loading, and in Fullscreen.

What to send with a report

Logging is always on. After a problem, **before** you start the game many more times, copy these from the `exmortis` folder:

1. `exmortis.log` and `exmortis.log.1` to `exmortis.log.5`. A new log starts at every launch and only five old ones are kept.
2. `bootstrap.log`. If it is missing, the newest `exmortis-bootstrap-*.log` in your `%TEMP%` folder.
3. `exmortis.ini`.
4. From `diagnostics\`: the folder of the problem (a crash folder is named `crash-...`) and `exits.log`. If you are not sure, send the whole folder.
5. For a desync: the `state-...-desync-...` files with their `.spans.txt` and `.inputs.txt`. Ask your opponent for theirs too.
6. For a replay problem: the `.nbreplay` file.

Put them in a zip and add:

- the time it happened,
- what you were doing, and whether you were playing or watching,

- the window title,
- a screenshot or a short video if it helps.

These files can hold Steam ids, player names and game memory. Send them privately to the person who looks into it.

18. Known issues in this beta

- **Practice rooms with three or more players** changed late in this build and had little testing. Tell us if Leave, the refill options or the rotation behave oddly there.

19. Credits

- **Aquatic**: main developer and maintainer of ExMortis.
- **Armonté and the AH3 mod developers**: their mod was the model for much of this one, and they helped during development.
- **The GekkoNet developers** (Jamie Meyer and contributors): GekkoNet is the rollback netcode library ExMortis is built on.

ExMortis also contains the work below. The full licence texts are in section 20.

Part	Used for	Licence
GekkoNet	Rollback netcode	BSD 2-Clause
zpp::bits	Packing data inside GekkoNet	MIT
SafetyHook	Attaching the mod's code to the game	Boost Software License 1.0
Zydis and Zycore	Reading the game's instructions, as part of SafetyHook	MIT
HookManager (WistfulHopes, in Armonté's fork)	Keeping the mod's attachments in order	MIT
HDE32, from MinHook	Measuring the game's instructions	BSD 2-Clause
xxHash	Check values	BSD 2-Clause
Dear ImGui	Tools used while developing the mod	MIT
Noto Sans JP	The text on the mod's own pages	SIL Open Font License 1.1

20. Licences

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MinHook and HDE32

```
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xxHash

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